

# STAT 4005 - Time Series

## Chapter 1 - Introduction

# Agenda

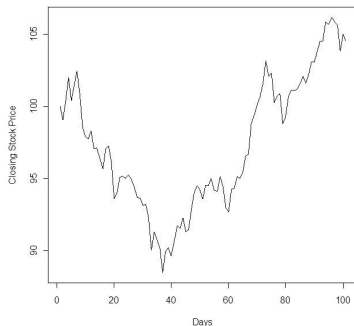
- 1 Basic description
- 2 Decomposition of time series
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    - Filtering
    - Differencing
  - Seasonal cycles
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    - Seasonal differencing
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# Basic description

## What is Time Series?

- A collection of values  $X_t : t = 1, 2, \dots, n$ 
  - $t$  : time index (e.g. days, months, years)
  - $X_t$  : observed data at different time (e.g. temperature, stock price)

Example of Time Series :



# Distinctive feature of time series

- In classical setting (e.g. t-test, regression), we assume  $X_t$ 's are independent
- In time series, we relax the independence assumption
  - Assume  $X_t$ 's are dependent
  - The dependence in time series is quantified by **auto-covariance**

For small  $k$ ,  $\text{Cov}(X_t, X_{t+k}) \neq 0$

# Some nonstandard features of $X_t$

## 1) **Unequally spaced data** ( $t = 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 15, \dots$ )

- Security Price are not available during non-trading days

## 2) **Continuous time series** ( $t$ can be any real number)

- Physical phenomenon (e.g. temperature)
  - We have  $\{X(t), t \in \mathbb{R}\}$  instead of  $\{X(t)\}_{t=1,2,\dots}$

## 3) **Aggregation**

- Data observed : an accumulation of some quantities over a period of time
- Examples:
  - Daily return = aggregation of tick by tick return
  - Daily rainfall = aggregation of hourly rainfall

# Some nonstandard features of $X_t$

## 4) Replicated time series

- Data observed : repeated measurement of same quantity across different subjects
- e.g.:  $X_t^{(j)}$  total weekly spending for the  $j$ -th person at week  $t$  where  $t = 1, 2, \dots, n$  are "repeated measurement" of subject  $j$
- $X_{t_1}^{(j)}$  and  $X_{t_2}^{(k)}$  are independent for all  $t_1, t_2$  if  $j \neq k$

## 5) Multiple time series

- $\mathbf{X}_t$  is a vector rather than a scalar
- e.g. Returns of a portfolio at time  $t$  :  $\mathbf{X}_t = (X_{1,t}, X_{2,t}, \dots, X_{p,t})$   
 $X_{i,t}$  : return of equity  $i$  at time  $t$
- Need to consider the "cross correlation" between different series  
e.g.  $\text{Cov}(X_{1,t}, X_{2,h})$

# Importance of time series

Example 1 - Using correlation to find a "better" confidence interval of  $\mu$

Model :  $X_t = \mu + a_t - \theta a_{t-1}$  where  $a_t \stackrel{\text{iid}}{\sim} N(0, 1)$

- $E(X_t) = \mu + E(a_t) - \theta E(a_{t-1}) = \mu$
- There exists time dependence because

$$\text{Cov}(X_t, X_{t-k}) = \begin{cases} -\theta & k = \pm 1 \\ 1 + \theta^2 & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

# Importance of time series

- Approximate 95% confidence interval for the mean  $\mu$

$$\bar{X} \pm 2\sqrt{\text{Var}(\bar{X})}$$

- Proof: By Central Limit Theorem (C.L.T.),

$$\frac{\bar{X} - \mu}{\sqrt{\text{Var}(\bar{X})}} \rightarrow N(0, 1)$$

$$\therefore P\left(-2 \leq \frac{\bar{X} - \mu}{\sqrt{\text{Var}(\bar{X})}} \leq 2\right) \approx 0.95$$

$$\Rightarrow P\left(\bar{X} - 2\sqrt{\text{Var}(\bar{X})} \leq \mu \leq \bar{X} + 2\sqrt{\text{Var}(\bar{X})}\right) \approx 0.95$$



# Importance of time series

Note that

$$\begin{aligned}\text{Var}(\bar{X}) &= \text{Cov}\left(\frac{1}{n} \sum_{t=1}^n X_t, \frac{1}{n} \sum_{t=1}^n X_t\right) = \frac{1}{n^2} \sum_{t=1}^n \sum_{s=1}^n \text{Cov}(X_t, X_s) \\&= \frac{1}{n^2} (n\text{Cov}(X_1, X_1) + 2(n-1)\text{Cov}(X_2, X_1)) \\&= \frac{1}{n^2} [n(1 + \theta^2) - 2(n-1)\theta] \\&= \frac{1}{n} \left[ (1 - \theta)^2 + \frac{2\theta}{n} \right].\end{aligned}$$

# Importance of time series

Therefore, a 95% C.I. is

$$\bar{X} \pm \frac{2}{\sqrt{n}} \left[ (1 - \theta)^2 + \frac{2\theta}{n} \right]^{1/2}.$$

- Without dependence ( $\theta = 0$ ), 95% C.I. is  $\bar{X} \pm \frac{2}{\sqrt{n}}$
- Need to incorporate correlation structure to produce a correct C.I.

Ratio of length of C.I.s is:

$$L(\theta) = \left\{ \frac{2}{\sqrt{n}} \left[ (1 - \theta)^2 + \frac{2\theta}{n} \right]^{1/2} \right\} / \left\{ \frac{2}{\sqrt{n}} \right\} = \left[ (1 - \theta)^2 + \frac{2\theta}{n} \right]^{1/2}$$

# Importance of time series

| $\theta$ | $L(\theta), n = 50$ |
|----------|---------------------|
| -1       | 1.99                |
| -0.5     | 1.49                |
| 0        | 1                   |
| 0.5      | 0.52                |
| 1        | 0.20                |

## Implication:

- If  $\theta = 1$ , but we wrongly assumed independence ( $\theta = 0$ ), then our C.I. is "too big"
- If  $\theta = -1$ , but we wrongly assumed independence ( $\theta = 0$ ), then our C.I. is "too small"
- Need to incorporate correlation structure to produce a correct C.I.

## Exercise

Model :  $X_t = \mu + Z_t - 0.5Z_{t-1}$  where  $Z_t \stackrel{\text{iid}}{\sim} N(0, 1)$ . Let  $(X_1, X_2, X_3, X_4, X_5) = (4, 5, 6, 4, 1)$ .

a Find  $\bar{X}$ .

$$\bar{X} = \frac{4+5+6+4+1}{5} = 4.$$

b Find the autocovariance function  $\gamma(k) = \text{Cov}(X_t, X_{t-k})$ .

$$\text{Cov}(X_t, X_{t-k}) = \begin{cases} -0.5 & k = \pm 1 \\ 1.25 & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

c Find  $\text{Var}(\bar{X})$ .

$$\text{Var}(\bar{X}) = \frac{1}{5} \left[ (1 - 0.5)^2 + \frac{2 \cdot 0.5}{5} \right] = 0.09$$

d Find the 95% C.I. for  $\mu$ .

$$(4 - 2 \cdot \sqrt{0.09}, 4 + 2 \cdot \sqrt{0.09}) = (3.4, 4.6)$$

# Exercise

Suppose that  $X_t = 10 + Z_t + 0.3Z_{t-1}$ ,  $Z_t \stackrel{\text{iid}}{\sim} N(0, 1)$ .

- (a) Find  $\text{Var}(X_t)$ ,  $\text{Cov}(X_t, X_{t+1})$ ,  $\text{Corr}(X_t, X_{t-1})$  and  $\text{Cov}(X_t, X_{t+k})$  for  $|k| \geq 2$ .
- (b) Let  $\bar{X} = \frac{1}{100} \sum_{t=1}^{100} X_t$ . Find  $\text{Var}(\bar{X})$ .
- (c) Give an example (a value of  $\bar{X}$ ) such that for the hypothesis  $H_0 : E(X_t) = 10$  verses  $H_1 : E(X_t) \neq 10$ , the tests
  - 1) accounting for time dependence, and
  - 2) without accounting for time dependence,give different conclusions.

a)

$$\begin{aligned}\text{Var}(X_t) &= \text{Var}(Z_t) + (0.3)^2\text{Var}(Z_{t-1}) \\ &= 1.09,\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_t, X_{t+1}) &= \text{Cov}(Z_t + 0.3Z_{t-1}, Z_{t+1} + 0.3Z_t) \\ &= \text{Cov}(Z_t, 0.3Z_t) \\ &= 0.3,\end{aligned}$$

$$\begin{aligned}\text{Corr}(X_t, X_{t-1}) &= \text{Cov}(X_t, X_{t-1})/(\sqrt{\text{Var}(X_t)}\sqrt{\text{Var}(X_{t-1})}) \\ &= \text{Cov}(Z_t + 0.3Z_{t-1}, Z_{t-1} + 0.3Z_{t-2})/\text{Var}(X_t) \\ &= \text{Cov}(0.3Z_{t-1}, Z_{t-1})/\text{Var}(X_t) \\ &= 0.275,\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_t, X_{t+k}) &= \text{Cov}(Z_t + 0.3Z_{t-1}, Z_{t+k} + 0.3Z_{t+k-1}) \\ &= 0.\end{aligned}$$

## Solution II

- Observe that  $\text{Var}(X_t)$ ,  $\text{Cov}(X_t, X_{t-1})$  are independent of  $t$ , and  $\text{Cov}(X_t, X_{t-k}) = 0$  for  $k \geq 2$ . So

$$\begin{aligned}\text{Var}\left(\frac{1}{n} \sum_{t=1}^n X_t\right) &= \frac{1}{n^2} \left( \sum_{t=1}^n \text{Var}(X_t) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \right) \\&= \frac{1}{n^2} \left( \sum_{t=1}^n \text{Var}(X_t) + 2 \sum_{i=1}^{n-1} \text{Cov}(X_t, X_{t-1}) \right) \\&= \frac{1}{n^2} (n \times 1.09 + 2(n-1) \times 0.3) \\&= 0.01684,\end{aligned}$$

where  $n = 100$ .

## Solution III

- Ⓔ) Without assuming dependence, we would have  $X_t = 10 + Z_t$  and thus

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{t=1}^n \text{Var}(Z_t) = \frac{1}{n} = 0.01.$$

For example, if  $\bar{X} = 9.78$ , without accounting for the dependence, the CI is

$$\bar{X} \pm 2\sqrt{\text{Var}(\bar{X})} = 9.78 \pm 2 \times \sqrt{0.01} = (9.58, 9.98),$$

which does not cover 10, thus  $H_0$  is rejected. However, when the dependence is accounted for, the CI is

$$\bar{X} \pm 2\sqrt{\text{Var}(\bar{X})} = 9.78 \pm 2 \times \sqrt{0.01684} = (9.52, 10.04),$$

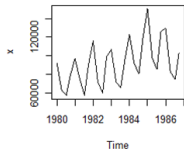
which covers 10, thus  $H_0$  is not rejected.



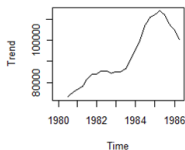
# Decomposition of time series

$$X_t = \underbrace{T_t}_{\text{(Trend)}} + \underbrace{S_t}_{\text{(Seasonality)}} + \underbrace{N_t}_{\text{(Noise)}}$$

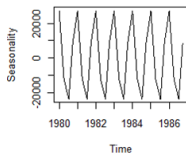
- $T_t + S_t \Rightarrow$  Macroscopic Component
- $N_t \Rightarrow$  Microscopic Component



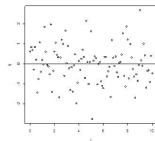
Series



Trend



Seasonality

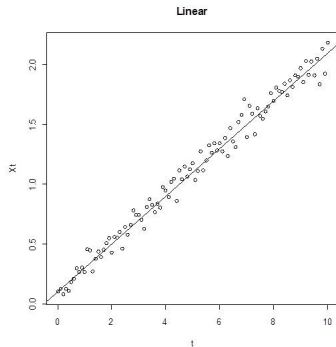


Noise

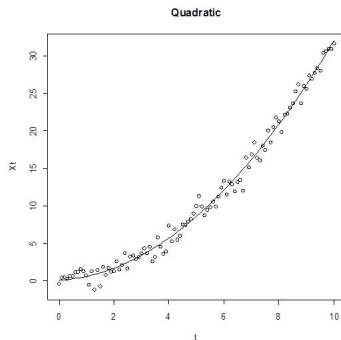
# Trends

## Examples

- Linear Trend :  $T_t = \alpha + \beta t$
- Quadratic Trend :  $T_t = \alpha + \beta t + \gamma t^2$



Linear Trend



Quadratic Trend

# Estimation of trends without seasonality

When  $X_t = T_t + N_t$  (no seasonal effect), the trend can be estimated by

## 1) Least Squares Method

- minimizes  $\sum (X_t - T_t)^2$
- $T_t$  is a simple model of trend, e.g.  $T_t = \alpha + \beta t$

## 2) Filtering

- $\hat{T}_t = S_m(X_t)$
- $S_m$  is a smoother: an operator that computes weighted average of observations near  $X_t$

## 3) Differencing

- $\Delta X_t = X_t - X_{t-1}$
- Removing trend instead of estimating trend

# 1. Least squares method

Idea : Find  $\alpha$  ,  $\beta_j$ s in  $T_t = \alpha + \beta_1 t + \cdots + \beta_k t^k$  such that

$$RSS = \sum_{t=1}^n (X_t - T_t)^2 \text{ is minimized}$$

Method: Regression  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e} \Rightarrow \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$

Comparing  $X_t = \alpha + \beta_1 t + \cdots + \beta_k t^k + N_t$  to  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$

$$\Rightarrow \mathbf{Y} = (X_1, \dots, X_n)'; \mathbf{X} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 2 & \cdots & 2^k \\ \vdots & & \vdots & \\ 1 & n & \cdots & n^k \end{pmatrix}$$

$$\Rightarrow \hat{T} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

# 1. Least squares method

R-Implementation:

```
n=100
k=2
Y=3+2*(1:n)+10*rnorm(n)
X=rep(1,n)
for (j in 1:k) {
  X=cbind(X,(1:n)^(j))
}
Trend.coef=solve(t(X)%*%X,t(X)%*%Y)
Trend=X%*%Trend.coef
ts.plot(Y)
points(1:n,Trend,col=2,type='l')
```

Drawbacks

- Only allow simple form of  $T_t$ , otherwise the minimization is difficult

# 1. Least squares method:

## Exercise:

- Given the data  $\{1.2, 2.1, 2.9, 3.8\}$ , estimate the trend in the form  $T_t = \alpha + \beta t$ .

$$\mathbf{Y} = \begin{pmatrix} 1.2 \\ 2.1 \\ 2.9 \\ 3.8 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix},$$

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}, \quad (\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{pmatrix},$$

$$\begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \begin{pmatrix} 0.35 \\ 0.86 \end{pmatrix}.$$

## 2. Filtering

- Idea : Smooth the series using **local data** to estimate the trend

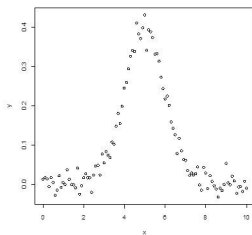
$$\underbrace{\hat{T}_t = S_m(X_t)}_{\text{"smoothed" series}} = \underbrace{\sum_{r=-q}^s a_r X_{t+r}}_{\text{Weighted average of } \{X_{t-q}, X_{t-q+1}, \dots, X_{t+s}\}}$$

- Local:  $q$  and  $s$  are much smaller comparing to the sample size  $n$ .
- The weight  $\{a_r\}$  are usually assumed to be symmetric ( $a_r = a_{-r}$ ) and normalized ( $\sum a_r = 1$ )

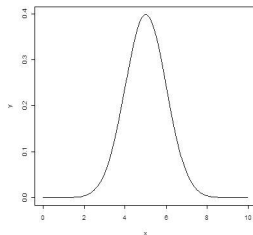
## 2. Filtering

- Low-pass filter: Filter out high-frequencies (volatile) signals  
Retains low-frequencies (smooth) signals

Low-Pass e.g.  $S_m(X_t) = \frac{\sum_{r=-q}^q X_{t+r}}{2q+1}$



before Low-pass



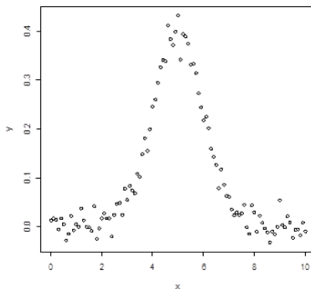
after Low-pass



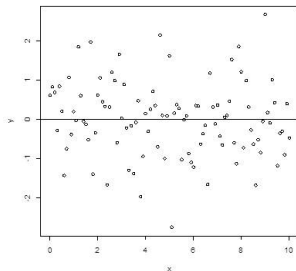
## 2. Filtering

- High-pass filter: Filter out low-frequencies (smooth) signals  
Retains high-frequencies (volatile) signals

High-Pass e.g.  $X_t - S_m(X_t)$



before High-pass



after High-pass

## 2. Filtering

### Examples

#### 1) Moving Average Filter

$$\hat{T}_t = \frac{1}{2q+1} \sum_{r=-q}^q X_{t+r}$$

- What is  $\{\hat{T}_t\}$  if  $\{X_t\} = \{1.1, 2.2, 2.7, 4.1, 5.2, 5.8\}$  and  $q = 1$ ?
- If  $X_t = \alpha + \beta t + N_t$  (Trend+Noise),

$$\begin{aligned} \hat{T}_t &= S_m(X_t) = \frac{1}{2q+1} \sum_{r=-q}^q [\alpha + \beta(t+r) + N_{t+r}] \\ &\approx \alpha + \beta t \quad \left( \text{smoothing: } \frac{1}{2q+1} \sum_{r=-q}^q N_{t+r} \approx 0 \right) \end{aligned}$$

## 2. Filtering

### Examples

#### 2) Spencer 15-point filter:

$$(a_0, a_1, \dots, a_7) = \frac{1}{320} (74, 67, 46, 21, 3, -5, -6, -3), \quad a_r = a_{-r}$$

- Property: Does not distort a cubic trend:
  - If  $X_t = T_t + N_t$ , where  $T_t = at^3 + bt^2 + ct + d$ , then

$$\begin{aligned}\hat{T}_t = S_m(X_t) &= \sum_{r=-7}^7 a_r T_{t+r} + \sum_{r=-7}^7 a_r N_{t+r} \\ &= at^3 + bt^2 + ct + d + \underbrace{\sum_{r=-7}^7 a_r N_{t+r}} \\ &\approx T_t \quad \left( \sum_{r=-7}^7 a_r N_{t+r} \approx 0 \right)\end{aligned}$$

- We say: the cubic trend "passes through" the filter

## 2. Filtering

Exercise: Spencer 15-point filter:

$$(a_0, a_1, \dots, a_7) = \frac{1}{320} (74, 67, 46, 21, 3, -5, -6, -3), \quad a_r = a_{-r}$$

What is the filtered/smoothed series if the input is

- $X_t = ct$

$$\begin{aligned} \frac{1}{320} [ & (-3) \cdot (-7) + (-6) \cdot (-6) + (-5) \cdot (-5) + 3 \cdot (-4) \\ & + 21 \cdot (-3) + 46 \cdot (-2) + 67 \cdot (-1) + 67 \cdot 1 + 46 \cdot 2 \\ & + 21 \cdot 3 + 3 \cdot 4 + (-5) \cdot 5 + (-6) \cdot 6 + (-3) \cdot 7 ] = 0. \end{aligned}$$

## 2. Filtering

- $X_t = bt^2$ ?

$$\begin{aligned} \frac{1}{320} [ & (-3) \cdot (-7)^2 + (-6) \cdot (-6)^2 + (-5) \cdot (-5)^2 + 3 \cdot (-4)^2 \\ & + 21 \cdot (-3)^2 + 46 \cdot (-2)^2 + 67 \cdot (-1)^2 + 67 \cdot 1^2 + 46 \cdot 2^2 \\ & + 21 \cdot 3^2 + 3 \cdot 4^2 + (-5) \cdot 5^2 + (-6) \cdot 6^2 + (-3) \cdot 7^2 ] = 0. \end{aligned}$$

- $X_t = at^3$ ?

$$\begin{aligned} \frac{1}{320} [ & (-3) \cdot (-7)^3 + (-6) \cdot (-6)^3 + (-5) \cdot (-5)^3 + 3 \cdot (-4)^3 \\ & + 21 \cdot (-3)^3 + 46 \cdot (-2)^3 + 67 \cdot (-1)^3 + 67 \cdot 1^3 + 46 \cdot 2^3 \\ & + 21 \cdot 3^3 + 3 \cdot 4^3 + (-5) \cdot 5^3 + (-6) \cdot 6^3 + (-3) \cdot 7^3 ] = 0 \end{aligned}$$

## 2. Filtering

### Theorem 1

A  $k^{th}$  order polynomial passes through a filter  $T_t$ , i.e.

$$\hat{T}_t = \sum_{r=-s}^s a_r T_{t+r} = T_t,$$

for  $T_t = c_0 + c_1 t + \cdots + c_k t^k$  if and only if

- ①  $\sum_{r=-s}^s a_r = 1$
- ②  $\sum_{r=-s}^s r^j a_r = 0$  for  $j = 1, 2, \dots, k$

## 2. Filtering

Application: Design a filter such that a quadratic trend would not be distorted.

We design a filter which has order  $s = 2$  does not distort a quadratic trend as follows,

$$a_r = a_{-r}$$

$$a_r = (a_0 = 1/4, a_1 = 1/2, a_2 = -1/8)$$

satisfy conditions

$$\sum_r a_r = 1, \quad \sum_r r a_r = 0, \quad \sum_r r^2 a_r = 0.$$

any other filter with different weights  $\{a_r\}$  are acceptable.

# Exercise

Let  $\{a_{-2}, a_{-1}, a_0, a_1, a_2\}$  be a filter and  $\{x_1, x_2, \dots, x_{10}\}$  be the observations.

- a) If the filter is applied to the observations, what is the length of the filtered sequence?
- b) Write, in terms of  $a_k$  and  $x_k$ s, the filtered value for the 6-th observation ( $x_6$ ).
- c) Suppose that  $a_0 = 0.5$ ,  $a_1 = 0.2$ ,  $a_2 = 0.05$  and the filter is symmetric. Does the filter pass through a linear trend without distortion? Does the filter passes through a quadratic trend without distortion?
- d) If  $a_0 = 0.5$ ,  $a_1 = a_{-1} = a_2 = a_{-2} = 0.1$ , do you think this filter is useful? Explain.



# Solution

a) The length of sequence is 6.

b)

$$\hat{x}_6 = a_{-2}x_4 + a_{-1}x_5 + a_0x_6 + a_1x_7 + a_2x_8.$$

c) The filter can pass any linear trend since

$$\sum_{r=-2}^2 a_r = 1,$$

$$\sum_{r=-2}^2 r a_r = 0.$$

But since

$$\sum_{r=-2}^2 r^2 a_r \neq 0,$$

the filter does not pass through quadratic trend.

d) The filter is not useful because  $\sum_{r=-2}^2 a_r = 0.9 \neq 1$ .

### 3. Differencing

- Differencing

$$\text{First Order:} \quad \Delta X_t = X_t - X_{t-1}$$

$$\text{Second Order:} \quad \Delta^2 X_t = \Delta(\Delta X_t)$$

- Definition: Backshift operator (B):

- $BX_t = X_{t-1}$

- $B^k X_t = X_{t-k}, k = 1, 2, \dots$

- Definition: Differencing operator ( $\Delta$ ):

- $\Delta X_t = (1 - B)X_t$

- $\Delta^k X_t = (1 - B)^k X_t, k = 1, 2, \dots$

- Exercise: What is  $\{\Delta X_t\}$  if  $\{X_t\} = \{1.1, 2.2, 2.7, 4.1, 5.2, 5.8\}$

$$\{\Delta X_t\} = \{1.1, 0.5, 1.4, 1.1, 0.6\}.$$

### 3. Differencing removes trend

- If  $X_t = \alpha + \beta t$ ,
  - $\Delta X_t = X_t - X_{t-1} = \alpha + \beta t - [\alpha + \beta(t-1)] = \beta$  (no Trend!)
- If  $X_t = \gamma t^p$ ,
  - $\Delta X_t = X_t - X_{t-1} = \gamma t^p - \gamma(t-1)^p = p\gamma t^{p-1} - C_2^p \gamma t^{p-2} + \dots$
- In general, if  $X_t = T_t + N_t$  and  $T_t = \sum_{j=0}^p a_j t^j$ ,
  - $\Delta^p X_t = p! a_p + \Delta^p N_t$
  - If the trend is a  $j^{th}$  degree polynomial, then the trend can be eliminated in differencing  $j$  times.

# Example

```
n=100
t=1:n
x=5-2*t+3*t^2-4*t^3+10*rnorm(n)
d1=diff(x)
d2=diff(d1)
d3=diff(d2)
d4=diff(d3)
d5=diff(d4)
par(mfrow=c(2,3))
ts.plot(x)
ts.plot(d1)
ts.plot(d2)
ts.plot(d3)
ts.plot(d4)
ts.plot(d5)
```

# Seasonal cycles

- General Decomposition

$$X_t = T_t + S_t + N_t$$

- Multiplicative Seasonal Component

$$X_t = T_t S_t N_t$$

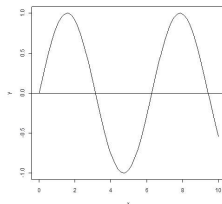
- Seasonal component  $S_t$ : period= $d$

- Requirements

- ①  $S_{t+d} = S_t$  (period  $d$ : repeating itself after time  $d$ )

- ②  $\sum_{j=1}^d S_j = 0$  (common effect is explained by  $T_t$ )

- e.g. season:  $d=4$ , month:  $d=12$ , week:  $d=7$



# Estimating/Removing seasonal effect

Difficulties when both  $T_t$  and  $S_t$  exist:

- Need to separate the effect of trend  $T_t$  and seasonal effect  $S_t$
- Simple model  $T_t = \sum_{j=0}^k a_j t^j$  does not work.
  - ⇒ Cannot use least squares method

Available methods:

- 1) Moving average method
- 2) Seasonal differencing

# 1. Moving average method

- STEP 1: Estimate the trend  $T_t$  by a special moving average filter
  - the moving average filter must be of length  $d$
  - ⇒ the estimated trend is free from seasonal effect because  $\sum_{j=1}^d S_j = 0$

$$\hat{T}_t = \begin{cases} \frac{1}{d} \left( \frac{1}{2}X_{t-q} + X_{t-q+1} + \dots + \frac{1}{2}X_{t+q} \right) & , \quad d = 2q \\ \frac{1}{d} \sum_{r=-q}^q X_{t+r} & , \quad d = 2q + 1 \end{cases}$$

# 1. Moving average method

- STEP 2: Given sample from  $t = 1, \dots, nd$ , where  $n$  is the number of complete cycle and  $d$  is the number of seasonal components, we estimate the seasonal components  $\hat{S}_j$ ,  $j = 1, \dots, d$ , as follows.

$$D_t = X_t - \hat{T}_t,$$

$$\overline{D} = \frac{1}{nd} \sum_{t=1}^{nd} D_t,$$

$$\hat{S}_j = \frac{1}{n} \sum_{k=0}^{n-1} (D_{kd+j} - \overline{D}).$$

- STEP 3: Use any filter for the series  $X_t - \hat{S}_t$  to get an improved  $\tilde{T}_t$  (can set  $\tilde{T}_t = \hat{T}_t$  if you are satisfied with the filter in Step 1.)
- RESULT:  $X_t = \tilde{T}_t + \hat{S}_t + \hat{N}_t$ , ( $\hat{N}_t = X_t - \tilde{T}_t - \hat{S}_t$ )



## 2. Seasonal differencing

- Seasonal differencing

$$\begin{aligned}\Delta^{(d)} X_t &= (1 - B^d) X_t \\ &= X_t - X_{t-d}\end{aligned}$$

- Seasonal differencing removes seasonal effects:

If  $X_t = S_t + N_t$  and period  $= d$ ,

then  $\Delta^{(d)} X_t = S_t - S_{t-d} + N_t - N_{t-d} = N_t - N_{t-d}$ .

(Recall  $S_t = S_{t-d}$ )

- Drawback - Lose  $d$  data points

- data  $= (X_1, X_2, \dots, X_n)$
- differenced data  $= (\Delta^{(d)} X_{d+1}, \dots, \Delta^{(d)} X_n)$

- No estimated seasonal effect  $\hat{S}_t$  is obtained.

# Example in R

- Data :  
Quarterly operating revenues of Washington power company 80-86  
<http://www.sta.cuhk.edu.hk/NHCHAN/TSBook2nd/dataset.html>
- Step 0: Time Series Plot

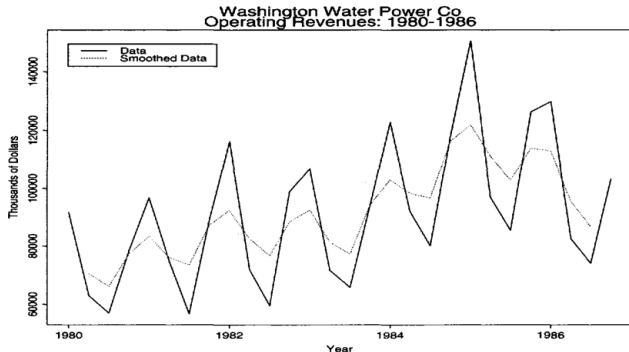


Fig. 1.2 Time series plots.

# Step 0: Time series plot

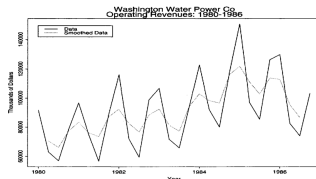


Fig. 1.2 Time series plots.

- Observations

- 1) Slightly increasing trend
- 2) Annual Cycle

- Lower in summer
- Higher in winter (heating?)

## R-Codes for reading data and time series plot

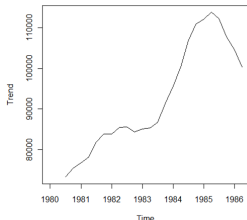
```
x=read.delim("washpower.dat",head=FALSE)
x=ts(x, frequency = 4, start = c(1980, 1))
ts.plot(x,main="Wasington Water Power Co")
```

# Decomposition $X_t = T_t + S_t + N_t$ : Step 1

- Step 1:

Estimate the trend  $\hat{T}_t$  by a filter running over one complete season ( $d = 4$ )

$$\begin{aligned}\hat{T}_3 &= \frac{\frac{1}{2}X_1 + X_2 + X_3 + X_4 + \frac{1}{2}X_5}{4} \\ \hat{T}_4 &= \frac{\frac{1}{2}X_2 + X_3 + X_4 + X_5 + \frac{1}{2}X_6}{4} \\ &\dots \\ \hat{T}_{26} &= \frac{\frac{1}{2}X_{24} + X_{25} + X_{26} + X_{27} + \frac{1}{2}X_{28}}{4}\end{aligned}$$



```
n=length(x)
# make empty vector to store trend
T1=rep(NA,n)
filter=c(0.125,0.25,0.25,0.25,0.125)
# compute filtered values recursively
radius=2;start=3;end=n-2
for ( k in start:end) {
  T1[k]=filter%*%x[(k-radius):(k+radius)]
}
ts.plot(T1,ylab="Trend")
```

# Decomposition $X_t = T_t + S_t + N_t$ : Step 2

- Step 2: Estimate seasonal effect from the trend-removed series  $D_t$

$$D_t = X_t - \hat{T}_t, \quad \bar{D} = \frac{1}{24} \sum_{i=3}^{26} D_i$$

Estimating the seasonal part  $S_i$

$$\hat{S}_1 = [(D_5 - \bar{D}) + (D_9 - \bar{D}) + \dots + (D_{25} - \bar{D})]/6$$

$$\hat{S}_2 =$$

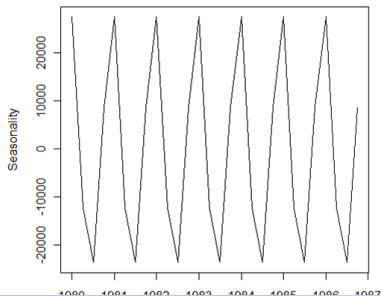
$$[(D_6 - \bar{D}) + (D_{10} - \bar{D}) + \dots + (D_{26} - \bar{D})]/6$$

$$\hat{S}_3 = [(D_3 - \bar{D}) + (D_7 - \bar{D}) + \dots + (D_{23} - \bar{D})]/6$$

$$\hat{S}_4 = [(D_4 - \bar{D}) + (D_8 - \bar{D}) + \dots + (D_{24} - \bar{D})]/6$$

$$\Rightarrow S_{i+4j} = S_i, \quad \text{all } i, j$$

```
d=4;           # set period=4
D=x-T1;
# na.rm=T means remove NA entries
D.bar=mean(D,na.rm=T)
# data of same season put in same col.
S.mat=matrix(D-D.bar,ncol=d,byrow=T)
# compute column(2) mean
S=apply(S.mat,2,mean,na.rm=T)
# make S the same length as x
S=rep(S,n/d)
ts.plot(S,ylab="Seasonality")
```



# Decomposition $X_t = T_t + S_t + N_t$ : Step 3

- Step 3: Re-estimate the trend from deseasonalized data  $Q_t = X_t - S_t$   
Re-estimating the trend by **any** reasonable filter, e.g.

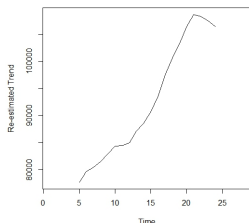
$$\tilde{T}_5 = \frac{1}{9}(Q_1 + \cdots + Q_9)$$

$$\tilde{T}_6 = \frac{1}{9}(Q_2 + \cdots + Q_{10})$$

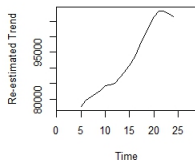
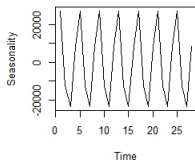
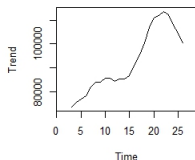
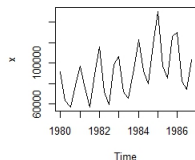
$$\cdots = \cdots$$

$$\tilde{T}_{24} = \frac{1}{9}(Q_{20} + \cdots + Q_{28})$$

```
Q=x-S
# make empty vector to store new trend
newT=rep(NA,n)
filter=rep(1/9,9)
# compute new filtered values recursively
radius=4;start=5;end=n-4
for ( k in start:end) {
  newT[k]=filter*%Q[(k-radius):(k+radius)]
}
ts.plot(newT,ylab="Re-estimated Trend")
```



# Putting four graphs together



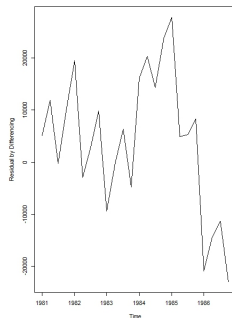
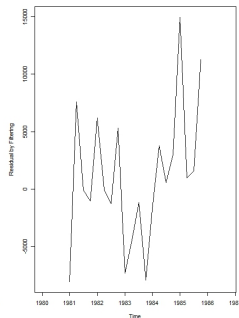
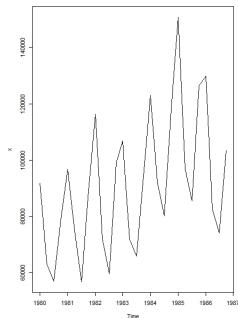
```
par(mfrow=c(2,2)) # make 2x2 plots
ts.plot(x)
ts.plot(T1,ylab="Trend")
ts.plot(S,ylab="Seasonality")
ts.plot(newT,ylab="Re-estimated Trend")
```

- Note that the re-estimated trend is smoother than the trend in Step 1.
- (Again) No need to do Step 3 if satisfied with the trend in Step 1.

# Example: Seasonal differencing

$$Y_t = (1 - B^4)X_t = X_t - X_{t-4}$$

```
y=diff(x,4)
par(mfrow=c(1,3))
ts.plot(x)
ts.plot(x-newT-S,ylab="Residual by Filtering")
ts.plot(y,ylab="Residual by Differencing")
```





# Decomposition $X_t = \widetilde{T}_t + \widehat{S}_t + \widehat{N}_t$

What to do after decomposition or differencing?

- Check the residual

- $\widehat{N}_t = X_t - \widetilde{T}_t - \widehat{S}_t$  or
- $\widehat{N}_t = X_t - X_{t-d}$

to detect further structure

- Sophisticated models for the microscopic structure in  $\widehat{N}_t$  will be discussed in later chapters

## Exercise

Use the following R-code to generate a series with trend, quarterly seasonal component and noise:

```
set.seed(4005)
season.effect=c(-20,-10,60,-30)
sea.com=rep(season.effect,20)
t=1:80
trend=-2+0.3*t+0.025*(t^2)
noise=rnorm(80,0,10)
data=trend+sea.com+noise
ts.plot(data)
```

Decompose the series data into three components. Compare the true and the estimated components.

# Summary of Chapter 1

## 1) Time Series

- Observations with time dependent structure
- $\text{Cov}(X_t, X_{t+k}) \neq 0$

## 2) Decomposition

$$X_t = \underbrace{T_t}_{(\text{Trend})} + \underbrace{S_t}_{(\text{Seasonality})} + \underbrace{N_t}_{(\text{Noise})}$$

### i) Removing Trend only

- Least Squares Method: Minimizes  $\sum_{t=1}^n (X_t - T_t)^2$
- Filters:  $\hat{T}_t = S_m(X_t) = \sum_{r=-q}^s a_r X_{t+r}$
- Differencing  $\Delta X_t = (1 - B)X_t = X_t - X_{t-1}$

### ii) Removing Trend and Seasonality

- Seasonal Filtering:
  - Step 1: Moving average filter with length  $d$  to obtain  $\hat{T}_t$
  - Step 2: Estimate  $\hat{S}_t$  from  $X_t - \hat{T}_t$
  - Step 3: Re-estimate  $\tilde{T}_t$  from  $X_t - \hat{S}_t$
- Seasonal Differencing  $\Delta^{(d)} X_t = (1 - B^d)X_t = X_t - X_{t-d}$

# STAT 4005 - Time Series

## **Chapter 2 - Probability Models**

# Agenda

## 1 Introduction

## 2 Stochastic process

- Definition 1 – Stochastic process
- Definition 2 – Finite dimensional distribution function
- Definition 3 – Strictly stationary
- Definition 4 – Weakly stationary
- Definition 5 – Autocovariance function

## 3 Strictly vs weakly stationary

## 4 Examples

## 5 Sample correlation function

## 6 Summary

- Assume that we have done the preliminary analysis to estimate trend  $T_t$  and seasonality  $S_t$
- What remains unexplained is the noise  $N_t$ :

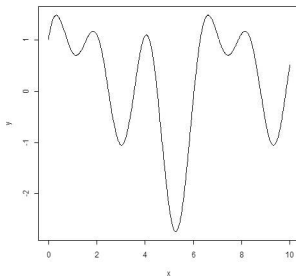
$$N_t = X_t - T_t - S_t$$

- We propose statistical models to describe the noise  $\{N_t\}$  to extract more information

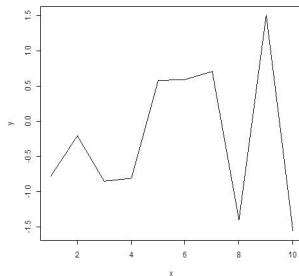
# Definition 1 – Stochastic process

- A collection of random variable  $\{X_t : t \in \mathcal{T}\}$  is called a stochastic process, where  $\mathcal{T}$  is an index set of time.
- Continuous  
 $\{X(t) : t \in [0, \infty)\}$
- Discrete  
 $\{X_t : t \in \{1, 2, \dots, n\}\}$

Continuous Time Process



Discrete Time Process



# Definition 1 – Stochastic process

Understand the meaning of **stochastic** (Random):

- $\{X_t : t = 1, 2, \dots, n\}$  can be regarded as a function of time  $t$
- We assume that there is a sample space  $\Omega$  (a set of all possible scenario) that **generates** the randomness
  - $\Omega = (\omega_1, \omega_2, \omega_3, \dots)$  (an  $\infty$ -face die)  
 $\Rightarrow \omega_1 \rightarrow \{X_t(\omega_1)\}; \omega_2 \rightarrow \{X_t(\omega_2)\}$  (each face gives one path)
- For a fixed  $\omega$ ,  $\{X_t(\omega)\}$  is called a sample function/ realization/ sample path
- For a fixed  $t$ ,  $X_t(\cdot)$  is a random variable
- In practice, we can only observe one sample path (i.e. the situation is generated by one particular  $\omega$ )
- To summarize important features from the data, we require the process to be repeating itself in some way (stationarity).



## Definition 2 – Finite dimensional distribution function

- Finite dimensional distribution functions:

$$F_{\mathbf{t}}(\mathbf{x}) = P(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n)$$

for all possible  $\mathbf{t} = (t_1, t_2, \dots, t_n)$  where,  $1 \leq t_1 < t_2 < \dots < t_n \leq \infty$  and  $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n, n = 1, 2, \dots$

## Definition 3 – Strictly stationary

- Strictly stationary - distribution of a process does not change over time.
- $\{X_t\}$  is said to be strictly stationary if
  - for all  $n$ ,
  - for all  $(t_1, t_2, \dots, t_n)$ ,
  - for all  $h$ ,

$$(X_{t_1}, \dots, X_{t_n}) \stackrel{d}{=} (X_{t_1+h}, \dots, X_{t_n+h}) ,$$

where " $\stackrel{d}{=}$ " means "equal in distribution", i.e.,

$$F_{\mathbf{t}}(\mathbf{x}) = P(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n) = P(X_{t_1+h} \leq x_1, \dots, X_{t_n+h} \leq x_n) = F_{\mathbf{t}+h}(\mathbf{x})$$

- May not be easy to check in practice because joint distribution  $F_{\mathbf{t}}(\mathbf{x})$  is difficult to compute.

## Definition 4 – Weakly stationary

- Weakly Stationary:
  - same mean
  - same variance and same covarianceacross time.
- Formally,  $\{X_t\}$  is weakly stationary/ second order stationary/ wide-sense stationary if
  - 1)  $E(X_t) = \mu$
  - 2)  $\text{Cov}(X_t, X_{t+h}) = \gamma(h)$  for all  $t$  and  $h$ .  
(auto-covariance only depends on time lag  $h$  , but not time  $t$ )
- Easier to verify weakly stationary than strictly stationary

## Definition 5 – Autocovariance function

- Autocovariance function (ACVF)

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$

- Autocorrelation function (ACF)

$$\begin{aligned}\rho(h) &= \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} \\ &= \frac{\gamma(h)}{\gamma(0)}.\end{aligned}$$

Under stationarity, we have the following properties:

- $\gamma(0) = \text{Var}(X_t) = \text{Var}(X_{t+h})$
- $\gamma(-h) = \gamma(h)$ .
- $\rho(-h) = \rho(h)$ .

## Exercise

Let  $X_t = (-1)^t Z$ , where  $Z$  is a random variable. Give necessary condition(s) on  $Z$  so that  $X_t$  is weakly stationary.

Solution:

$$\begin{aligned} E(X_t) &= (-1)^t E(Z), \\ \text{Cov}(X_t, X_{t+k}) &= (-1)^{2t+k} \text{Cov}(Z, Z) = (-1)^k \text{Var}(Z). \end{aligned}$$

Note that the autocovariance function is independent of  $t$ . To make the mean constant we need  $(-1)^t E(Z)$  to be independent of  $t$ , which implies  $E(Z) = -E(Z)$ , i.e.  $E(Z) = 0$ . Therefore, we require  $E(Z) = 0$  if  $\{X_t\}$  is stationary.

# Strictly vs weakly stationary

- Strictly: distribution of the process unchanged.  
Weakly: mean/covariance unchanged.
- Strictly stationary implies weakly stationary whenever  $E(X_t^2) < \infty$ .
- Weakly stationary implies strictly stationary when  $\{X_t\}$  is normal, but in general not true.
- In the subsequent chapters, unless indicated specifically, stationary always refers to weakly stationary.

# Strictly vs weakly stationary

- Basic statistics: If  $X \sim t_{df}$ , then  $E(X^k) < \infty$  if and only if  $k < df$ .
1.  $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} t_2$ 
    - $\{X_t\}$  is strictly stationary
    - But  $E(X_t^2) = \infty$ ,  
 $\therefore \{X_t\}$  is **NOT** weakly stationary
  2.  $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} t_3$ 
    - Now  $E(X_t^2) < \infty$ ,  
 $\therefore \{X_t\}$  is strictly and weakly stationary
  3.  $X_1 \sim \text{Exp}(1) - 1, X_2, X_3, \dots \stackrel{\text{iid}}{\sim} N(0, 1)$ 
    - $\{X_t\}$  is weakly stationary:  
 $E(X_t) = 0, \text{Var}(X_t) = 1, \text{Cov}(X_t, X_{t+k}) = 0$ , all  $t$  and  $k \neq 0$
    - But  $\{X_t\}$  is **NOT** strictly stationary, since  $X_1 \stackrel{d}{\neq} X_2$ .

## Exercise

Let  $W_t = (-1)^t Y_t$  where  $Y_t$  are independently and identically distributed with  $P(Y_t = k) = 0.2, k = -2, -1, 0, 1, 2$  for all integers  $t$ .

- (a) Is  $W_t$  strictly stationary?
- (b) Is  $W_t$  weakly stationary?

Solution:

- (a) Note that  $P(-Y_t = k) = 0.2, k = -2, -1, 0, 1, 2$  for all integers  $t$ . Thus  $Y_t$  and  $-Y_t$  have the same distribution. Since  $\{Y_t\}_{t=1,\dots}$  are independent, the series  $\{(-1)^t Y_t\}_{t=1,\dots}$  are actually independent and identically distributed. Thus,  $\{W_t\}$  is strictly stationary.
- (b) Since

$$E(W_t^2) = E(Y_t^2) = 0.2((-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2) = 2 < \infty,$$

the second moment of  $W_t$  exists. Thus the strict stationarity of  $\{W_t\}$  implies its weak stationarity.



## Exercise

Give an example of a series  $\{Z_t\}$  with a constant mean and  $\text{Corr}(Z_t, Z_{t+k})$  being free of  $t$ , but  $\{Z_t\}$  is not stationary.

Solution: Consider  $Z_t = ta_t$ , where  $a_t \stackrel{\text{iid}}{\sim} N(0, 1)$ . By straightforward algebra,

$$\begin{aligned}E(Z_t) &= tE(a_t) = 0, \\ \text{Var}(Z_t) &= t^2 \text{Var}(a_t) = t^2, \\ \text{Cov}(Z_t, Z_{t+k}) &= t(t+k) \text{Cov}(a_t, a_{t+k}) = 0, \quad \text{for } k \neq 0, \\ \text{Corr}(Z_t, Z_{t+k}) &= \frac{t(t+k) \text{Cov}(a_t, a_{t+k})}{\sqrt{t^2(t+k)^2}} = \begin{cases} 1, & \text{for } k = 0, \\ 0, & \text{for } k \neq 0, \end{cases}\end{aligned}$$

Clearly,  $E(Z_t)$  and  $\text{Corr}(Z_t, Z_{t+k})$  are independent of time  $t$ . But  $\{Z_t\}$  is not stationary since the variance  $\text{Var}(Z_t) = t^2$  depends on time.

## Example 1: White Noise

If a time series  $\{X_t\}$  is a sequence of uncorrelated random variables with

$$\rho(h) = \begin{cases} 1, & h = 0, \\ 0, & |h| \geq 1, \end{cases} \quad (1)$$

with zero mean and constant variance, then it is called a **white noise** sequence.

- i.i.d. random variables are white noise if their second moments exist
- white noise *may not be* i.i.d.  
(e.g.,  $X_1 \sim \text{Exp}(1) - 1$ ,  $X_k \sim N(0, 1)$ ,  $k \geq 2$ )
- independence is a stronger assumption than zero-correlation

## Exercise

Show that  $Z_t = a_t a_{t-2}$  is a white noise process, where  $a_t$  are i.i.d. sequence of random variables with zero mean and variance  $\sigma_a^2$ .

Solution: Note that  $E(Z_t) = E(a_t a_{t-2}) = 0$ . In addition, for  $k \neq 0$ ,

$$\begin{aligned}\text{Cov}(Z_t, Z_{t+k}) &= E(Z_t Z_{t+k}) - E(Z_t)E(Z_{t+k}) \\ &= E(a_t a_{t-2} a_{t+k} a_{t+k-2}) - 0 \\ &= 0 \\ \text{Var}(Z_t) &= E(Z_t^2) \\ &= E(a_t^2)E(a_{t-2}^2) \\ &= \sigma_a^4 < \infty.\end{aligned}$$

Thus, the mean of  $\{Z_t\}$  is constant, and the autocorrelation function is the same as that of a white noise process, so  $Z_t \sim WN(0, \sigma_a^4)$ .

## Example 2: Same Observation

If  $\{Y_t\}$  satisfies  $Y_1 = Y_2 = \dots = Y$  with  $E(Y^2) < \infty$ , then

- $\rho(h) = \text{Corr}(Y, Y) = 1$  for all  $h$ .
- Essentially there is only one observation.
- In comparison with White Noise in Example 1, we have

$$\begin{aligned}\frac{1}{n}(X_1 + \dots + X_n) &\rightarrow E(X_1) = \mu, \\ \frac{1}{n}(Y_1 + \dots + Y_n) &= Y.\end{aligned}$$

There is as much randomness in  $n$ -th sample average as there is in the first observation  $Y_1$ .

## Example 3: Periodic Function

- Suppose a time series  $\{X_t\}$  satisfies

$$X_t = A \cos \theta t + B \sin \theta t,$$

where  $A$  and  $B$  are i.i.d. random variables with mean 0 and variance  $\sigma^2$ .

- $\{X_t\}$  is weakly stationary because
  - $E(X_t) = E(A) \cos \theta t + E(B) \sin \theta t = 0$
  -

$$\begin{aligned} & \text{Cov}(X_{t+h}, X_t) \\ &= \text{Cov}(A \cos \theta(t+h) + B \sin \theta(t+h), A \cos \theta t + B \sin \theta t) \\ &= \cos \theta t \cos \theta(t+h) \text{Var}(A) + \sin \theta t \sin \theta(t+h) \text{Var}(B) \\ &= \sigma^2 \cos(\theta(t+h) - \theta t) \quad (\cos(A-B) = \cos A \cos B + \sin A \sin B) \\ &= \sigma^2 \cos \theta h, \end{aligned}$$

independent of  $t$ .

## Example 4: Moving Average MA(1) Model

Let  $\{\epsilon_t\}$  be a white noise sequence with  $\text{Var}(\epsilon_t) = 1$  and  $X_t = \epsilon_t + 0.5\epsilon_{t-1}$ . Is  $\{X_t\}$  stationary? Find the ACVF and ACF of  $\{X_t\}$ .

$$E(X_t) = 0.$$

$$\gamma(h) = \begin{cases} 1.25, & h = 0, \\ 0.5, & h = 1, \\ 0, & h \geq 2. \end{cases}$$

$$\rho(h) = \begin{cases} 1, & h = 0, \\ 0.4, & h = 1, \\ 0, & h \geq 2. \end{cases}$$

Since  $E(X_t)$  and  $\rho(h)$  are independent of  $t$ ,  $\{X_t\}$  is weakly stationary.

## Exercise

Suppose that  $\{a_t\}_{t=1,\dots}$  are independent and identically distributed random variables with mean 0 and finite variance  $\sigma_a^2$ , and  $\theta_i$  are constants for  $i = 0, 1, 2, 3, 4$ . If

$$X_t = -\theta_0 a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2} - \theta_3 a_{t-3} - \theta_4 a_{t-4},$$

then compute the autocorrelation function  $\rho_k$  of  $X_t$  for  $k = 0, 1, 2, \dots$

Solution:

$$\text{Var}(X_t) = \text{Var}(-\theta_0 a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2} - \theta_3 a_{t-3} - \theta_4 a_{t-4})$$

$$= (\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+1}) = (\theta_0 \theta_1 + \theta_1 \theta_2 + \theta_2 \theta_3 + \theta_3 \theta_4) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+2}) = (\theta_0 \theta_2 + \theta_1 \theta_3 + \theta_2 \theta_4) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+3}) = (\theta_0 \theta_3 + \theta_1 \theta_4) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+4}) = (\theta_0 \theta_4) \sigma_a^2.$$

## Exercise (Continued)

Thus, the autocorrelation function is

$$\begin{aligned}\rho(0) &= 1 \\ \rho(1) &= \frac{\theta_0\theta_1 + \theta_1\theta_2 + \theta_2\theta_3 + \theta_3\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(2) &= \frac{\theta_0\theta_2 + \theta_1\theta_3 + \theta_2\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(3) &= \frac{\theta_0\theta_3 + \theta_1\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(4) &= \frac{\theta_0\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(k) &= 0, \quad |k| > 4\end{aligned}$$



# Exercise

Consider the process  $Z_t = a_t - 1.7a_{t-1} + 0.72a_{t-2}$  where  $\{a_t\}$  are i.i.d. with finite variance  $\sigma_a^2 = 4$ .

- (1) Find  $\text{Var}(Z_t)$ .
- (2) Find the autocovariance function of the process.
- (3) Find  $\text{Var}(\frac{1}{15} \sum_{t=1}^{15} Z_t)$ .

# Solution

(1)

$$\begin{aligned}\text{Var}(Z_t) &= \text{Var}(a_t - 1.7a_{t-1} + 0.72a_{t-2}) \\ &= (1 + (1.7)^2 + (0.72)^2)\sigma_a^2 = 17.6\end{aligned}$$

(2)

$$\begin{aligned}\gamma(0) &= \text{Var}(Z_t) = 17.6 \\ \gamma(1) &= \text{Cov}(Z_t, Z_{t+1}) = -11.7 \\ \gamma(2) &= \text{Cov}(Z_t, Z_{t+2}) = 2.88 \\ \gamma(k) &= 0 \quad |k| > 2\end{aligned}$$

(3)

$$\begin{aligned}\text{Var}\left(\frac{1}{15}\sum_{t=1}^{15} Z_t\right) &= \frac{1}{15}\gamma(0) + \frac{1}{15^2}2(15-1)\gamma(1) + \frac{1}{15^2}2(15-2)\gamma(2) \\ &= 0.05.\end{aligned}$$

## Example 5: Moving Average $MA(\infty)$ Model

Let  $\{\epsilon_t\}$  be a white noise sequence with  $\text{Var}(\epsilon_t) = 1$  and  $X_t = \sum_{k=0}^{\infty} \psi_k \epsilon_{t-k}$  and  $\sum_{k=0}^{\infty} \psi_k^2 < \infty$ . Is  $\{X_t\}$  stationary? Find the ACVF and ACF of  $\{X_t\}$ .

$$E(X_t) = 0.$$

$$\gamma(h) = \begin{cases} \sum_{k=0}^{\infty} \psi_k^2, & h = 0, \\ \sum_{k=0}^{\infty} \psi_k \psi_{k+h}, & h \geq 1. \end{cases}$$

$$\rho(h) = \begin{cases} 1, & h = 0, \\ \frac{\sum_{k=0}^{\infty} \psi_k \psi_{k+h}}{\sum_{k=0}^{\infty} \psi_k^2}, & h \geq 1. \end{cases}$$

Note that by Cauchy-Schwarz inequality,

$$\sum_{k=0}^{\infty} |\psi_k \psi_{k+h}| \leq \sqrt{\left(\sum_{k=0}^{\infty} \psi_k^2\right) \left(\sum_{k=0}^{\infty} \psi_{k+h}^2\right)} < \infty,$$

hence the sum  $\sum_{k=0}^{\infty} \psi_k \psi_{k+h}$  is well defined. Since  $E(X_t)$  and  $\rho(h)$  are independent of  $t$ ,  $\{X_t\}$  is weakly stationary.

For the  $MA(\infty)$  model to be well defined, we usually require that  $\sum_{k=0}^{\infty} |\psi_k| < \infty$ . Such that the sum

$$X_t = \sum_{k=0}^{\infty} \psi_k \epsilon_{t-k}$$

converges absolutely. This is a stronger condition because

$$\sum_{k=0}^{\infty} \psi_k^2 \leq \left( \sum_{k=0}^{\infty} |\psi_k| \right)^2 < \infty.$$

# Sample ACVF and ACF

$$\text{ACVF} : \gamma(h) = \text{Cov}(X_t, X_{t+h})$$

$$\text{ACF} : \rho(h) = \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} = \frac{\gamma(h)}{\gamma(0)}$$

- Both ACVF and ACF are *population quantities*. It means that they involve the unknown true distribution of the process  $\{X_t\}$
- If we have a dataset  $X_1, X_2, \dots, X_n$  but we do not know the true probability distribution of the time series, we can estimate the ACF/ACVF by
  - Sample ACVF:  $C_h = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X})$
  - Sample ACF :  $r_h = C_h/C_0$

# Sample ACVF and ACF

- Sample ACVF:  $C_h = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X})$
- Why  $\frac{1}{n}$  but not  $\frac{1}{n-h}$ ?

- To ensure the **non-negative definiteness** of covariance matrix

$$\widehat{\text{Var}}((X_1, \dots, X_n)) = (C_{|i-j|})_{i,j \in \{1, \dots, n\}} = \frac{1}{n} M M',$$

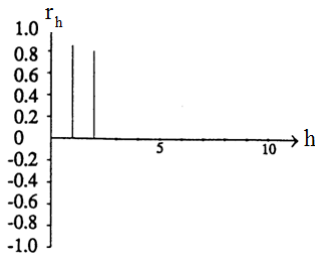
where with  $\tilde{X}_k = X_k - \bar{X}$ ,

$$M = \begin{pmatrix} \tilde{X}_1 & \tilde{X}_2 & \tilde{X}_3 & \cdots & \tilde{X}_n & 0 & 0 & 0 \\ 0 & \tilde{X}_1 & \tilde{X}_2 & & \tilde{X}_{n-1} & \tilde{X}_n & 0 & 0 \\ 0 & 0 & \tilde{X}_1 & & \tilde{X}_{n-2} & \tilde{X}_{n-1} & \tilde{X}_n & 0 \\ & & & \ddots & & & & \\ & & & & \tilde{X}_1 & \tilde{X}_2 & & \tilde{X}_n \end{pmatrix}.$$

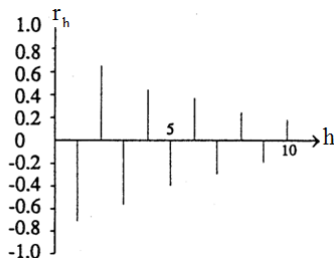
- Denote  $\mathbf{X} = (X_1, \dots, X_n)$ ,  
 $\widehat{\text{Var}}(\mathbf{a}'\mathbf{X}) = \mathbf{a}'\widehat{\text{Var}}(\mathbf{X})\mathbf{a} = \frac{1}{n}(\mathbf{a}'M)(\mathbf{a}'M)' \geq 0$  implies that  
 $(C_{|i-j|})_{i,j \in \{1, \dots, n\}}$  is non-negative definite.

# Correlogram - ACF plot

Correlogram - ACF plot: A graph to summarize  $r_h$  (`acf(x)` in R)



Example 1

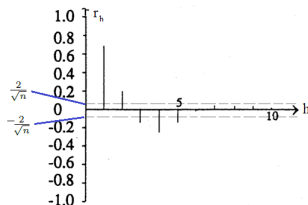


Example 2

- Trust  $r_h$  up to  $h = n/3$ , otherwise  $r_h$  is not accurate
- By definition,  $r_0 = \frac{C_0}{C_0} = 1$

# Sample Correlation Function

- Questions commonly of interest:
  - Is  $\text{Corr}(X_t, X_{t+h}) = 0$  ?
  - ⇔ Is  $r_h$  significantly different from 0 ?
- Solution:
  - Results: If  $\{X_t\}$  is white noise, then  $r_h \sim N(0, \frac{1}{n})$  for each  $h \geq 1$  when  $n$  is large.
  - Rigorous proof is out of the scope of this course, interested students may refer to Chapter 6 of the following book.
    - Brockwell, P. J., and Davis, R. A. (1991). Time series: theory and methods. Springer.
  - Draw horizontal lines at levels  $\pm \frac{2}{\sqrt{n}}$  on acf and check for exceedance.





## Example: ACF plot

Scratch the sample ACF plot for the dataset  $(2, 5, 3, 5, 6, 3, 4)$ . Is this time series a white noise sequence?

Since  $n = 7$ , the maximum value of  $h$  is the integer part of  $7/3$  or 2. We have

$$\overline{X} = 4,$$

$$C_0 = \frac{1}{7} \sum_{t=1}^7 (X_t - \overline{X})^2 = 1.714286,$$

$$C_1 = \frac{1}{7} \sum_{t=1}^6 (X_t - \overline{X})(X_{t+1} - \overline{X}) = -0.5714286,$$

$$C_2 = \frac{1}{7} \sum_{t=1}^5 (X_t - \overline{X})(X_{t+2} - \overline{X}) = 0,$$

$$r_0 = 1,$$

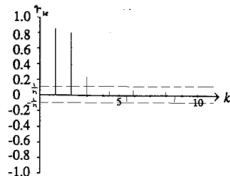
$$r_1 = \frac{C_1}{C_0} = -0.333333,$$

$$r_2 = 0.$$

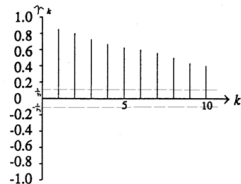
Since  $\frac{2}{\sqrt{n}} = 0.7559289$ , all the sample ACF are insignificant. We may conclude that the time series is a white noise sequence.

# Short memory vs long memory

- Short Memory = Short term correlation

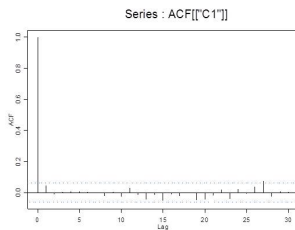


- Long Memory = long term correlation

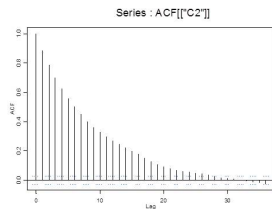


# More examples of ACF function

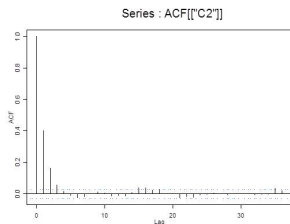
- $Y_t = Z_t, \{Z_t\}$  are i.i.d.



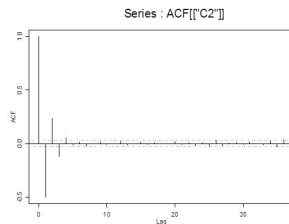
- $Y_t = 0.9Y_{t-1} + Z_t$



- $Y_t = 0.4Y_{t-1} + Z_t$

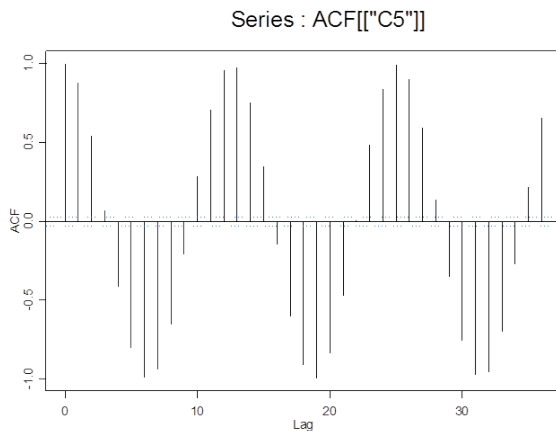


- $Y_t = -0.5Y_{t-1} + Z_t$



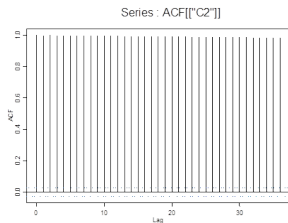
# More examples of ACF function

- $Y_t = a \cos(t\omega)$  , where  $a, \omega$  are constants

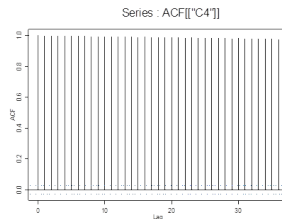


# More examples of ACF function

- $Y_t = Y_{t-1} + Z_t$



- $Y_t = at + Z_t$



1. In both cases, ACF decay very slowly
2. Commonly observed in **non-stationary** process
3. In practice, perform detrending/filtering and study the resulting stationary noises

## Remarks on sample ACF function

- Notice that the examples above suggest that we can "identify" a time series through inspection of its ACF.
- Although this sounds promising, it is not a procedure that is always free of error.
- When we calculate the ACF of any given series with a fixed sample size  $n$ , we cannot put too much confidence in the values of  $r_k$  for large value of  $k$ .
- Furthermore, if there is a trend in the data,

$$X_t = T_t + N_t.$$

then  $X_t$  becomes nonstationary and the idea of inspecting the ACF of  $X_t$  becomes questionable. It is therefore important to detrend the data before interpreting their ACF.

## Summary of Chapter 2

- Strictly Stationary: Joint distribution remain unchanged
- Weakly Stationary: Mean and covariance structure remain unchanged
- White Noise:  $E(X_t) = 0$ ,  $\text{Cov}(X_t, X_k) = \sigma^2 1_{\{t=k\}}$

$$\text{ACVF} : \gamma(h) = \text{Cov}(X_t, X_{t+h})$$

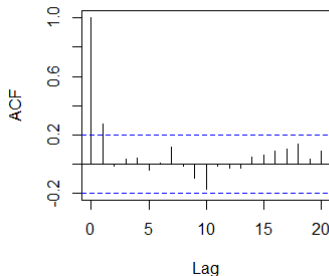
$$\text{ACF} : \rho(h) = \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} = \frac{\gamma(h)}{\gamma(0)}$$

$$\text{Sample ACVF} : C_h = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X})$$

$$\text{Sample ACF} : r_h = C_h / C_0$$

# Summary of Chapter 2

ACF plot (ACF  $r_h$  against lag  $h$ ):



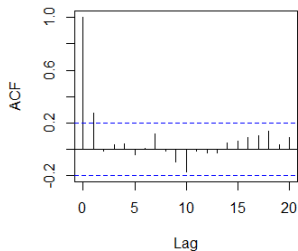
Check if  $\rho(h) = \text{Corr}(X_t, X_{t+h}) = 0$ :

- $\{X_t\}$  is white noise  $\Rightarrow r_h \sim N(0, \frac{1}{n})$  for all  $h \geq 1$
- Draw horizontal lines at levels  $\pm \frac{2}{\sqrt{n}}$  on acf and check for exceedance.

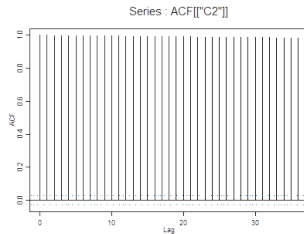


# Summary of Chapter 2

- Stationary



- Non-stationary



# STAT 4005 - Time Series

## Chapter 3 - ARMA Model

# Agenda

- 1 Introduction
- 2 Moving Average model (MA)
  - Stationarity
  - Invertibility
- 3 Autoregressive model (AR)
  - Stationarity
  - Asymptotic stationarity
  - Causality
  - Autocovariance of  $AR(p)$  model via Yule-Walker equation
- 4 ARMA model
- 5 ARIMA model
- 6 Seasonal model
- 7 Summary

# Introduction

- ARMA( $p, q$ ) model

- The most common model for **stationary** time series

$$\underbrace{Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p}}_{\text{Autoregressive (AR)}} = \underbrace{Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q}}_{\text{Moving Average (MA)}}$$

- $Y_t$  is observation,
- $Z_t \sim \text{WN}(0, \sigma^2)$  is white noise.
- Equivalently, the ARMA( $p, q$ ) model can be written as

$$\phi(B)Y_t = \theta(B)Z_t,$$

where

$$\begin{aligned}\phi(B) &= I - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p, \\ \theta(B) &= I - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q,\end{aligned}$$

are **characteristic polynomials** without common roots, i.e., no  $x$  s.t.  $\phi(x) = \theta(x) = 0$ , and  $B$  is the backshift operator.

- ARIMA( $p, d, q$ ) model

The most common model for **non-stationary** time series is

$$\phi(B) \underbrace{(1 - B)^d}_{\text{Integrated}} Y_t = \theta(B) Z_t.$$

- If  $Y_t$  follows an ARIMA( $p, d, q$ ) model, then  $\Delta^d Y_t = (1 - B)^d Y_t$  follows an ARMA( $p, q$ ) model, where  $(1 - B)^d Y_t$  is the  $d$ -times differenced series from  $Y_t$ .

# Differencing

For a time series  $Y_t$ ,

- When  $d = 1$ , the first order differenced series is  $\Delta Y_t = (1 - B)Y_t = Y_t - Y_{t-1}$ .
- When  $d = 2$ , the second order differenced series is

$$\begin{aligned}(1 - B)^2 Y_t &= \Delta^2 Y_t \\ &= \Delta(\Delta Y_t) \\ &= \Delta(Y_t - Y_{t-1}) \\ &= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) \\ &= Y_t - 2Y_{t-1} + Y_{t-2} \\ &= (1 - 2B + B^2)Y_t.\end{aligned}$$

- In general, the  $d$ -th order differenced series can be expressed as

$$(1 - B)^d Y_t = \sum_{k=0}^d \binom{d}{k} (-B)^k Y_t = \sum_{k=0}^d \binom{d}{k} (-1)^k Y_{t-k}.$$

# No common root

- Consider  $(1 - \psi B)Y_t = (1 - \psi B)Z_t$ , i.e.,

$$Y_t - \psi Y_{t-1} = Z_t - \psi Z_{t-1}, \quad Z_t \sim WN(0, \sigma^2). \quad (1)$$

- Now,  $\phi(B) = \theta(B) = (1 - \psi B)$   
 $\Rightarrow \phi\left(\frac{1}{\psi}\right) = \theta\left(\frac{1}{\psi}\right) = 0 \Rightarrow \frac{1}{\psi}$  is the common root.
- Apply (1) repeatedly, we have

$$Y_t - Z_t = \psi Y_{t-1} - \psi Z_{t-1} = \psi(Y_{t-1} - Z_{t-1}) = \cdots = \psi^k(Y_{t-k} - Z_{t-k})$$

for all  $k \geq 1$ .

- Thus, the only **stationary** solution for (1) is

$$Y_t = Z_t$$

for all  $t \geq 0$ .

- Remark:**  $Y_t = Z_t + \psi^t$  is a **non-stationary** solution for (1)

## Exercise

Identify the following as specific ARIMA( $p, d, q$ ) models

a)  $Y_t - 0.5Y_{t-1} + 0.06Y_{t-2} = Z_t + 0.6Z_{t-1} - 0.16Z_{t-2}$

b)  $Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} = Z_t + 0.2Z_{t-1}$

c)  $Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} = Z_t - 0.2Z_{t-1}$

Solution:

a)

$$Y_t - 0.5Y_{t-1} + 0.06Y_{t-2} = Z_t + 0.6Z_{t-1} - 0.16Z_{t-2}$$

$$(1 - 0.3B)(1 - 0.2B)Y_t = (1 + 0.8B)(1 - 0.2B)Z_t$$

$$(1 - 0.3B)Y_t = (1 + 0.8B)Z_t.$$

It is an ARIMA(1,0,1) model.



# Exercise

b)

$$\begin{aligned}Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} &= Z_t + 0.2Z_{t-1} \\(1 - B)(1 + 0.2B)Y_t &= (1 + 0.2B)Z_t \\(1 - B)Y_t &= Z_t.\end{aligned}$$

It is an ARIMA(0,1,0) model.

c)

$$\begin{aligned}Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} &= Z_t - 0.2Z_{t-1} \\(1 + 0.2B)(1 - B)Y_t &= (1 - 0.2B)Z_t\end{aligned}$$

It is an ARIMA(1,1,1) model.

# Moving Average Model

## Definition 1 (MA( $q$ ) model)

A stochastic process  $\{Y_t\}_{t=1,2,\dots}$  follows an MA( $q$ ) model if

$$Y_t = Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q}$$

where  $Z_t \sim WN(0, \sigma^2)$ .

- Sometimes, we further assume  $Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

# Moving Average Model: Stationarity

## Proposition 2 (Stationarity)

For all  $q$  and  $(\theta_1, \dots, \theta_q) \in \mathbb{R}^q$ , the  $MA(q)$  model  $Y_t = Z_t - \theta_1 Z_{t-1} - \dots - \theta_q Z_{t-q}$ ,  $Z_t \sim WN(0, \sigma^2)$  defines a stationary process.

## Proof.

Using the properties of white noise, i.e.  $E(Z_t) = 0$ ,  $\text{Cov}(Z_t, Z_k) = \sigma^2 1_{\{t=k\}}$ , we have

1.  $E(Y_t) = 0$
2.  $\text{Var}(Y_t) = (1 + \theta_1^2 + \dots + \theta_q^2) \sigma^2$
3.  $\text{Cov}(Y_t, Y_{t+k}) = \begin{cases} 0 & , \quad |k| > q \\ \sigma^2 \sum_{i=0}^{q-|k|} \theta_i \theta_{i+|k|} & , \quad |k| \leq q, (\theta_0 = -1) \end{cases}$

$\Rightarrow$  MA model is always stationary.



# Moving Average Model

## Example 3

Find the ACF of the MA(2) model:  $Y_t = Z_t + 0.6Z_{t-1} - 0.16Z_{t-2}$ ,  $Z_t \sim \text{W.N.}(0, \sigma^2)$ .

$$\gamma(0) = (1 + (-0.6)^2 + 0.16^2)\sigma^2 = 1.3856\sigma^2,$$

$$\gamma(1) = ((-1) \cdot (-0.6) + (-0.6) \cdot 0.16)\sigma^2 = 0.504\sigma^2,$$

$$\gamma(2) = (-1) \cdot 0.16\sigma^2 = -0.16\sigma^2,$$

$$\gamma(k) = 0 \text{ for } k \geq 3,$$

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = 0.3637,$$

$$\rho(2) = \frac{\gamma(2)}{\gamma(0)} = -0.1155,$$

$$\rho(k) = 0 \text{ for } k \geq 3,$$

$$\rho(k) = \rho(-k) \text{ for } k < 0.$$

# Moving Average Model

## Example 4

- Different MA models can have the same ACF:

A. MA(1):  $Y_t = Z_t - \theta Z_{t-1}$ ,  $Z_t \sim WN(0, 1)$ ,  
 $\gamma(0) = 1 + \theta^2$ ,  $\gamma(1) = -\theta$

$$\rho(k) = \begin{cases} 1 & , \quad k = 0 \\ -\frac{\theta}{1+\theta^2} & , \quad |k| = 1 \\ 0 & , \quad \text{otherwise} \end{cases}$$

B. MA(1):  $X_t = Z_t - \frac{1}{\theta} Z_{t-1}$ ,  $Z_t \sim WN(0, 1)$ ,  
 $\gamma(0) = 1 + 1/\theta^2$ ,  $\gamma(1) = -1/\theta$

$$\rho(k) = \begin{cases} 1 & , \quad k = 0 \\ -\frac{\theta}{1+\theta^2} & , \quad |k| = 1 \\ 0 & , \quad \text{otherwise} \end{cases}$$

# Moving Average Model - Invertibility

## Definition 5 (Invertibility)

A model is **invertible** if the white noise  $Z_t$  can be expressed as a linear combination of the past observations  $Y_t, Y_{t-1}, \dots$ , i.e.,

$$Z_t = \sum_{k=0}^{\infty} \psi_k Y_{t-k},$$

where  $\sum_{k=0}^{\infty} |\psi_k| < \infty$  (so that the summation is well defined).

- Example: MA(1) model:  $Y_t = Z_t - \theta Z_{t-1}$

$$\begin{aligned} Z_t &= Y_t + \theta Z_{t-1} = Y_t + \theta(Y_{t-1} + \theta Z_{t-2}) \\ &= Y_t + \theta Y_{t-1} + \theta^2 Z_{t-2} = Y_t + \theta Y_{t-1} + \theta^2(Y_{t-2} + \theta Z_{t-3}) \\ &= Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Z_{t-3} \\ &= Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \theta^4 Y_{t-4} + \dots \\ &= \sum_{k=0}^{\infty} \theta^k Y_{t-k}, \quad (\text{only if } |\theta| < 1) \end{aligned}$$

# Moving Average Model

- For MA(1) model  $Y_t = Z_t - \theta Z_{t-1}$  with  $|\theta| < 1$ , we have

$$Z_t = Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \theta^4 Y_{t-4} + \cdots \quad (2)$$

- Recall the formula  $(1 - x)^{-1} = 1 + x + x^2 + \cdots + x^k + \cdots$

- Proof:

$$\begin{aligned} & (1 - x)(1 + x + x^2 + \cdots) \\ &= (1 + x + x^2 + \cdots) - (x + x^2 + \cdots) = 1 \end{aligned}$$

- Write  $Y_t = Z_t - \theta Z_{t-1} = (1 - \theta B)Z_t$ , we "invert"  $(1 - \theta B)$  to get

$$\begin{aligned} Z_t &= (1 - \theta B)^{-1} Y_t \\ &= (1 + \theta B + \theta^2 B^2 + \theta^3 B^3 + \theta^4 B^4 + \cdots) Y_t \\ &= Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \theta^4 Y_{t-4} + \cdots \end{aligned}$$

- Same as (2)
- $Z_t = (1 - \theta B)^{-1} Y_t = \sum_{j=0}^{\infty} \theta^j Y_{t-j}$  should be understood as notation rather than mathematical operation.

# Moving Average Model - Model Selection

We have seen that  $\rho(k) = \begin{cases} 1 & , k = 0 \\ -\frac{\theta}{1+\theta^2} & , |k| = 1 \\ 0 & , \text{otherwise} \end{cases}$  is the ACF of either

$$\begin{aligned} \text{A. } Y_t &= Z_t - \theta Z_{t-1} = (1 - \theta B)Z_t \\ \Rightarrow Z_t &= Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \dots \end{aligned}$$

$$\begin{aligned} \text{B. } Y_t &= Z_t - \frac{1}{\theta} Z_{t-1} = (1 - \frac{1}{\theta} B)Z_t \\ \Rightarrow Z_t &= Y_t + \frac{1}{\theta} Y_{t-1} + \frac{1}{\theta^2} Y_{t-2} + \dots \end{aligned}$$

- Which model is more preferable?
  - If  $|\theta| < 1$  , series (B) does not converge,  
 $\Rightarrow$  (A) is invertible (more preferable)  
(B) is **NOT** invertible
  - If  $|\theta| > 1$  , series (A) does not converge,  
 $\Rightarrow$  (A) is **NOT** invertible  
(B) is invertible (more preferable)



# Invertibility for general MA Model

## Theorem 6 (Invertibility)

*An MA( $q$ ) process  $\{Y_t\}$  is invertible if the roots of the equation  $\theta(B) = 0$  all lie outside the unit circle*

## Proof.

Factorizing the characteristic polynomial yields

$$\begin{aligned}Y_t &= Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q} \\&= \theta(B) Z_t \\&= (1 - \bar{\theta}_1 B)(1 - \bar{\theta}_2 B) \cdots (1 - \bar{\theta}_q B) Z_t\end{aligned}$$

- $\frac{1}{\bar{\theta}_k}$ s are the roots of the equation  $\theta(B) = 0$
- $(1 - \bar{\theta}_k B)$  can be "inverted" to the other side if  $|\bar{\theta}_k| < 1$
- $\therefore$  The model is **invertible** if all  $|\bar{\theta}_k| < 1$ 
  - all  $|\bar{\theta}_k| < 1 \Leftrightarrow$  all  $\frac{1}{|\bar{\theta}_k|} > 1 \Leftrightarrow$  all roots lie outside the unit circle



## Exercise

Which of the following MA model(s) is(are) invertible?

- a)  $Y_t = Z_t + 0.6Z_{t-1} - 0.16Z_{t-2}$
- b)  $Y_t = Z_t - 0.6Z_{t-1} - 0.72Z_{t-2}$
- c)  $Y_t = Z_t - 0.5Z_{t-2} - 0.5Z_{t-4}$

Solution:

- a) The roots of the quadratic equation  $\theta(x) = 1 + 0.6x - 0.16x^2$  are  $x_1 = -1.25$  and  $x_2 = 5$ . Since  $|x_i| > 1$  for  $i = 1, 2$ , the model is invertible.
- b) The roots of the quadratic equation  $\theta(x) = 1 - 0.6x - 0.72x^2$  are  $x_1 = -1.6667$  and  $x_2 = 0.8333$ . Since  $|x_2| \leq 1$ , the model is not invertible.
- c) The roots of the quartic equation  $\theta(x) = 1 - 0.5x^2 - 0.5x^4 = (x^2 + 2)(0.5 - 0.5x^2)$  are  $x_1 = -\sqrt{2}i$ ,  $x_2 = \sqrt{2}i$ ,  $x_3 = -1$  and  $x_4 = 1$ . Since  $|x_3|, |x_4| \leq 1$ , the model is not invertible.

# Autoregressive Model

## Definition 7 (Autoregressive AR( $p$ ) Model)

A stochastic process  $\{Y_t\}$  follows an AR( $p$ ) model if it satisfies

$$\begin{aligned} Y_t &= \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t, \text{ or} \\ \phi(B)Y_t &= Z_t, \end{aligned}$$

where  $Z_t \sim \text{WN}(0, \sigma^2)$  is white noise and

$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$  is the characteristic polynomial.

AR model is very **popular** because

- Closely resembles traditional regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + z$$

$\therefore$  traditional theoretical results can be used.

- Intuitively, today's value depends on previous days.

## AR model with non-zero mean

$\{Y_t\}$  is an  $AR(p)$  process with mean  $\mu$  if it satisfies  $\phi(B)(Y_t - \mu) = Z_t$  for all  $t$

$$\begin{aligned}(Y_t - \mu) &= \phi_1(Y_{t-1} - \mu) + \cdots + \phi_p(Y_{t-p} - \mu) + Z_t \\ \phi(B)(Y_t - \mu) &= Z_t \\ \phi(B)Y_t &= \phi(1)\mu + Z_t,\end{aligned}$$

in which  $\phi(1) = 1 - \phi_1 - \phi_2 - \cdots - \phi_p$ .

## Exercise

Express the AR(2) model  $(1 - \phi_1 B - \phi_2 B^2)(Y_t - \mu) = Z_t$  in the form  $Y_t = \tilde{\phi}_0 + \tilde{\phi}_1 Y_{t-1} + \tilde{\phi}_2 Y_{t-2} + Z_t$ .

Solution:

$$\begin{aligned}(1 - \phi_1 B - \phi_2 B^2)(Y_t - \mu) &= Z_t, \\ Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} - \mu(1 - \phi_1 - \phi_2) &= Z_t,\end{aligned}$$

Hence,

$$Y_t = \mu(1 - \phi_1 - \phi_2) + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t.$$

# Exercise

Consider the stationary process  $Y_t = \mu + \phi Y_{t-1} + Z_t$ , where  $Z_t \sim WN(0, \sigma^2)$ , and  $|\phi| < 1$ . Find  $E(Y_t)$  and  $\text{Cov}(Y_t, Y_{t-k})$  for  $k = 0, 1, 2, \dots$

# Solution

From the assumed stationarity, we have  $E(Y_t) = E(Y_{t-1})$ , thus  $E(Y_t) = \mu + \phi E(Y_{t-1}) = \mu + \phi E(Y_t)$ , giving

$$E(Y_t) = \frac{\mu}{1 - \phi}.$$

Taking variance on both sides of  $Y_t = \mu + \phi Y_{t-1} + Z_t$ , and using the stationarity assumption  $\text{Var}(Y_t) = \text{Var}(Y_{t-1})$ , we have

$$\begin{aligned} \text{Var}(Y_t) &= \text{Var}(\mu + \phi Y_{t-1} + Z_t) = \phi^2 \text{Var}(Y_{t-1}) + \sigma^2 \\ \Rightarrow \text{Var}(Y_t) &= \gamma(0) = \frac{\sigma^2}{1 - \phi^2}. \end{aligned}$$

Considering the covariance of  $Y_t = \mu + \phi Y_{t-1} + Z_t$  with each of  $Y_{t-k}$ ,  $k = 1, 2, \dots$ , we have the autocovariance function  $\gamma(k)$ :

- $\gamma(1) = \text{Cov}(Y_t, Y_{t-1}) = \text{Cov}(\mu + \phi Y_{t-1} + Z_t, Y_{t-1}) = \phi \gamma(0) = \frac{\phi \sigma^2}{1 - \phi^2}.$
- For  $k \geq 2$ ,  $\gamma(k) = \text{Cov}(Y_t, Y_{t-k}) = \text{Cov}(\mu + \phi Y_{t-1} + Z_t, Y_{t-k})$   
 $= \phi \gamma(k-1) = \phi^2 \gamma(k-2) = \dots = \phi^k \gamma(0) = \frac{\phi^k \sigma^2}{1 - \phi^2}.$

# Stationarity

## Lemma 8 (MA( $\infty$ ): Stationarity)

If  $Y_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$ ,  $Z_t \sim WN(0, \sigma^2)$  and  $\sum_{i=0}^{\infty} |\psi_i| < \infty$ , then  $\{Y_t\}$  is a stationary process.

### Proof.

Using the properties of white noise, i.e.  $E(Z_t) = 0$ ,  $\text{Cov}(Z_t, Z_k) = \sigma^2 1_{\{t=k\}}$ , we have

1.  $E(Y_t) = 0$
  2.  $\text{Var}(Y_t) = \sigma^2 \sum_{i=0}^{\infty} \psi_i^2 \leq \sigma^2 (\sum_{i=0}^{\infty} |\psi_i|)^2 < \infty$
  3.  $\text{Cov}(Y_t, Y_{t+k}) = \sigma^2 \sum_{i=0}^{\infty} \psi_i \psi_{i+k}$  is independent of  $t$
- $\Rightarrow \{Y_t\}$  is stationary.





# AR(1) model: Stationarity

## Proposition 9 (Stationarity of AR(1) process)

*The process  $Y_t = \phi Y_{t-1} + Z_t$ ,  $Z_t \sim WN(0, \sigma^2)$  is stationary if  $|\phi| \neq 1$ .*

### Proof.

A) For  $|\phi| < 1$ , inverting the AR polynomial gives

$$(1 - \phi B)Y_t = Z_t \implies Y_t = (1 - \phi B)^{-1}Z_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}.$$

B) For  $|\phi| > 1$ , inverting a "modified" AR polynomial gives

$$\left(1 - \frac{1}{\phi B}\right)(-\phi B)Y_t = Z_t \implies (-\phi B)Y_t = \left(1 - \frac{1}{\phi B}\right)^{-1}Z_t = \sum_{k=0}^{\infty} \frac{1}{\phi^k} Z_{t+k}$$

$$\therefore Y_{t-1} = -\sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+k} \implies Y_t = -\sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+1+k}$$

- In both (A) and (B), the coefficients of the WN decay to 0 rapidly as  $k \rightarrow \infty$  and thus the **summations are well-defined**.
- Both (A) and (B) are stationary (check  $E(Y_t)$  and  $\text{Cov}(Y_t, Y_{t+k})$ ).

But (B) is **not** useful: It depends on future noises



## AR(1) model with $|\phi| > 1$

Note that when  $|\phi| > 1$ , the expression

$$Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$$

is not a valid representation since the sum is divergent for  $|\phi| \geq 1$ . However, if we consider

$$\begin{aligned} Z_t^* &= Y_t - \frac{1}{\phi} Y_{t-1} \\ &= Z_t + \left(\phi - \frac{1}{\phi}\right) Y_{t-1} \\ &= Z_t - \left(\phi - \frac{1}{\phi}\right) \sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+k} \\ &= \frac{1}{\phi^2} Z_t - \left(1 - \frac{1}{\phi^2}\right) \sum_{k=1}^{\infty} \frac{1}{\phi^k} Z_{t+k}. \end{aligned}$$

It can be shown that  $Z_t^*$  defined above is a white noise process with  $\text{Var}(Z_t^*) = \frac{\sigma^2}{\phi^2}$ .

## AR(1) model with $|\phi| > 1$

With the white noise process  $Z_t^*$ ,  $Y_t$  could be regarded as an AR(1) model

$$Y_t = \phi^* Y_{t-1} + Z_t^*$$

with  $\phi^* = \frac{1}{\phi}$  such that

$$|\phi^*| = \frac{1}{|\phi|} < 1.$$

$Y_t$  would have a causal representation with respect to  $Z_t^*$  as

$$Y_t = \sum_{k=0}^{\infty} \phi^{*k} Z_{t-k}^*.$$

Thus, for a stationary AR(1) model, we typically assume that  $|\phi| < 1$ .

# Example: Stationarity

## Example 10

Find a stationary solution for the AR(2) model

$$(1 - 0.3B)(1 - 1.2B)Y_t = Z_t.$$

$$(1 - 0.3B)(1 - 1.2B)Y_t = Z_t$$

$$\left(1 - \frac{1}{1.2B}\right)(-1.2B)Y_t = (1 - 0.3B)^{-1}Z_t$$

$$-1.2BY_t = \left(1 - \frac{1}{1.2B}\right)^{-1} \sum_{k=0}^{\infty} 0.3^k Z_{t-k}$$

$$Y_{t-1} = -\frac{1}{1.2} \sum_{l=0}^{\infty} \frac{1}{1.2^l} \sum_{k=0}^{\infty} 0.3^k Z_{t+l-k}$$

$$Y_t = -\sum_{l=0}^{\infty} \sum_{k=0}^{\infty} \frac{0.3^k}{1.2^{l+1}} Z_{t+1+l-k}$$

## Theorem 11 (Stationarity of AR( $p$ ) process)

*The process  $\phi(B)Y_t = Z_t$ ,  $Z_t \sim WN(0, \sigma^2)$  is stationary if no root of  $\phi(x)$  is on the unit circle.*

## Exercise

Consider an  $MA(\infty)$  process  $Y_t = Z_t + C(Z_{t-1} + Z_{t-2} + \dots)$ , where  $C$  is a fixed constant,  $Z_t \sim WN(0, \sigma^2)$ .

- (a) Is  $Y_t$  weakly stationary?
- (b) Let  $W_t = Y_t - Y_{t-1}$ . Show that  $\{W_t\}$  is a stationary  $MA(1)$  model.
- (c) Find the autocorrelation function of  $\{W_t\}$ .

Solution:

- (a) From the definition of  $Y_t$  we have

$$\text{Var}(Y_t) = [1 + C^2(1 + 1 + 1 + \dots)]\sigma^2 = \infty,$$

so  $Y_t$  is not weakly stationary.

## Exercise

- (b) Since  $W_t = Y_t - Y_{t-1} = Z_t + (C - 1)Z_{t-1}$ , we have  $E(W_t) = 0$  and autocovariance function  $\gamma(k) = \text{Cov}(W_t, W_{t-k})$  satisfying

$$\begin{aligned}\gamma(0) &= (1 + (C - 1)^2)\sigma^2 \\ \gamma(1) &= (C - 1)\sigma^2 \\ \gamma(k) &= 0 \quad \text{for } |k| > 1.\end{aligned}$$

Thus,  $\{W_t\}$  is a stationary MA(1) process.

- (c) From the above autocovariance function, we get the autocorrelation function

$$\begin{aligned}\rho(0) &= 1 \\ \rho(1) &= \frac{C - 1}{1 + (C - 1)^2} \\ \rho(k) &= 0 \quad \text{for } |k| > 1.\end{aligned}$$

# Asymptotic stationary

Suppose an AR(1) process does not go back to the remote past, but starts from an initial value  $Y_0$ ,

$$Y_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \cdots + \phi^{t-1} Z_1 + \phi^t Y_0$$

- $Y_0$  is usually fixed but **NOT** distributed as  $\sum_{k=0}^{\infty} \phi^k Z_{-k}$
- The process  $Y_t$  is not stationary because:

$$\begin{aligned} E(Y_t) &= \phi^t E(Y_0) \\ \text{Var}(Y_t) &= \sigma^2 \left( 1 + \phi^2 + \cdots + \phi^{2(t-1)} \right) + \phi^{2t} \text{Var}(Y_0) \end{aligned}$$

are time dependent (depends on  $t$ ).

- However, as  $t \rightarrow \infty$ ,  $E(Y_t) \rightarrow 0$  and  $\text{Var}(Y_t) \rightarrow \frac{\sigma^2}{1-\phi^2}$

$\therefore$  we say that  $\{Y_t\}$  is asymptotic stationary!

# Causal condition

## Definition 12 (Causality)

$\{Y_t\}$  is **causal** if it can be expressed as a linear combination of the **past white noise**  $Z_t, Z_{t-1}, \dots$ , i.e.,

$$Y_t = \sum_{k=0}^{\infty} \psi_k Z_{t-k},$$

where  $\sum_{k=0}^{\infty} |\psi_k| < \infty$  (so that the summation is well defined).

## Example 13 (AR(1) Model)

For AR(1) model  $Y_t = \phi Y_{t-1} + Z_t$ ,  $Z_t \sim WN(0, \sigma^2)$ , we have shown that

- If  $|\phi| < 1$ , then  $Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$   
 $\Rightarrow \{Y_t\}$  is causal (Note that  $\sum_{k=0}^{\infty} |\psi_k| = \sum_{k=0}^{\infty} |\phi^k| = \frac{1}{1-|\phi|} < \infty$ )
- If  $|\phi| > 1$ , then  $Y_t = -\sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+1+k}$   
 $\Rightarrow \{Y_t\}$  is **NOT** causal (It depends on future noises)



# Remarks on causality and stationarity

What is the difference between stationarity and causality for AR(1) ?

- Causality: depends on past data

$|\phi| < 1$  :  $Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$  is causal

$|\phi| > 1$  :  $Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$  no longer exist  $\Rightarrow$  **NOT** causal

- Stationarity: constant mean and covariance over time

- $|\phi| < 1$  :

$Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$  is a stationary solution

- $|\phi| > 1$  :

$Y_t = \sum_{i=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+1+k}$  is a stationary solution but **NOT** causal

# Causality Theorem

## Theorem 14 (AR( $p$ ) Model)

An AR( $p$ ) process  $\phi(B)Y_t = Z_t$ ,  $Z_t \sim WN(0, \sigma^2)$  is causal if the roots of the characteristic polynomial

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$$

are outside the unit circle.

## Proof.

Factorization:  $\phi(B) = (1 - \xi_1 B)(1 - \xi_2 B) \dots (1 - \xi_p B)$

The roots of  $\phi(x)$  are  $x = \frac{1}{\xi_1}, \dots, \frac{1}{\xi_p}$ .

- The roots are outside unit circle  $\Leftrightarrow \left| \frac{1}{\xi_k} \right| > 1$ , for all  $k = 1, \dots, p$
- $\Leftrightarrow |\xi_k| < 1$ , for all  $k = 1, \dots, p$
- $\Leftrightarrow (1 - \xi_k B)$ , for all  $k = 1, \dots, p$ , can be "inverted" to another side and represented by past noises.
- $\Leftrightarrow \{Y_t\}$  is causal. □

# Exercise

Which of the following AR model(s) is(are) causal? Give the causal representation up to the second term.

a)  $Y_t + 0.6Y_{t-1} - 0.16Y_{t-2} = Z_t$

b)  $Y_t - 0.6Y_{t-1} - 0.72Y_{t-2} = Z_t$

c)  $Y_t - 0.5Y_{t-2} - 0.5Y_{t-4} = Z_t$

Solution:

- a) The roots of the quadratic equation  $\phi(x) = 1 + 0.6x - 0.16x^2$  are  $x_1 = -1.25$  and  $x_2 = 5$ . Since  $|x_i| > 1$  for  $i = 1, 2$ , the model is causal.

# Solution

To find the causal representation,

$$\begin{aligned}Y_t + 0.6Y_{t-1} - 0.16Y_{t-2} &= Z_t \\(1 + 0.8B)(1 - 0.2B)Y_t &= Z_t.\end{aligned}$$

Hence,

$$\begin{aligned}&Y_t \\&= (1 - 0.2B)^{-1}(1 + 0.8B)^{-1}Z_t \\&= \sum_{l=0}^{\infty} \sum_{k=0}^{\infty} (0.2)^l (-0.8)^k Z_{t-k-l} \\&= Z_t - 0.8Z_{t-1} + 0.2Z_{t-1} + (-0.8)^2 Z_{t-2} + 0.2^2 Z_{t-2} - 0.8 \cdot 0.2Z_{t-2} + \dots \\&= Z_t - 0.6Z_{t-1} + 0.52Z_{t-2} + \dots\end{aligned}$$

- b) The roots of the quadratic equation  $\phi(x) = 1 - 0.6x - 0.72x^2$  are  $x_1 = -1.6667$  and  $x_2 = 0.8333$ . Since  $|x_2| \leq 1$ , the model is not causal.
- c) The roots of the quartic equation  $\theta(x) = 1 - 0.5x^2 - 0.5x^4 = (x^2 + 2)(0.5 - 0.5x^2)$  are  $x_1 = -\sqrt{2}i$ ,  $x_2 = \sqrt{2}i$ ,  $x_3 = -1$  and  $x_4 = 1$ . Since  $|x_3|, |x_4| \leq 1$ , the model is not causal.

## Exercise

Find the AR and MA representations of the following process:

$$Y_t + 0.6Y_{t-1} = Z_t + 0.5Z_{t-1}, \quad Z_t \sim WN(0, \sigma^2).$$

# Solution

For AR representation, we have

$$\begin{aligned}(1 + 0.6B)Y_t &= (1 + 0.5B)Z_t \\ \Rightarrow Z_t &= (1 + 0.5B)^{-1}(1 + 0.6B)Y_t \\ &= \sum_{j=0}^{\infty} (-0.5)^j B^j (1 + 0.6B)Y_t \\ &= \left( \sum_{j=0}^{\infty} (-0.5)^j B^j + 0.6 \sum_{j=0}^{\infty} (-0.5)^j B^{j+1} \right) Y_t \\ &= \left( \sum_{j=0}^{\infty} (-0.5)^j B^j + 0.6 \sum_{j=1}^{\infty} (-0.5)^{j-1} B^j \right) Y_t \\ &= \left( 1 + 0.1 \sum_{j=1}^{\infty} (-0.5)^{j-1} B^j \right) Y_t \\ &= Y_t + 0.1 \sum_{j=1}^{\infty} (-0.5)^{j-1} Y_{t-j} .\end{aligned}$$

# Solution

Similarly, we have the MA representation

$$\begin{aligned}(1 + 0.6B)Y_t &= (1 + 0.5B)Z_t \\ \Rightarrow Y_t &= (1 + 0.6B)^{-1}(1 + 0.5B)Z_t \\ &= \sum_{j=0}^{\infty} (-0.6)^j B^j (1 + 0.5B)Z_t \\ &= \left( \sum_{j=0}^{\infty} (-0.6)^j B^j + 0.5 \sum_{j=0}^{\infty} (-0.6)^j B^{j+1} \right) Z_t \\ &= \left( \sum_{j=0}^{\infty} (-0.6)^j B^j + 0.5 \sum_{j=1}^{\infty} (-0.6)^{j-1} B^j \right) Z_t \\ &= \left( 1 - 0.1 \sum_{j=1}^{\infty} (-0.6)^{j-1} B^j \right) Z_t \\ &= Z_t - 0.1 \sum_{j=1}^{\infty} (-0.6)^{j-1} Z_{t-j} .\end{aligned}$$



# Exercise

Consider the AR(2) process

$$Y_t = 0.5Y_{t-1} - 0.06Y_{t-2} + Z_t.$$

Find the values of  $\psi_j$ ,  $j = 0, 1, 2, 3, \dots$  if the process is written in the form of general linear model

$$Y_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}.$$

# Solution

$$\begin{aligned}(1 - 0.5B + 0.06B^2)Y_t &= Z_t \\ \Rightarrow (1 - 0.2B)(1 - 0.3B)Y_t &= Z_t \\ \Rightarrow Y_t &= (1 - 0.2B)^{-1}(1 - 0.3B)^{-1}Z_t \\ &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} 0.2^j 0.3^k B^{j+k} Z_t \\ &= \sum_{m=0}^{\infty} \sum_{j,k \geq 0, j+k=m} (0.2^j 0.3^k) B^m Z_t \\ &= \sum_{m=0}^{\infty} \sum_{j=0}^m (0.2^j 0.3^{m-j}) B^m Z_t \\ &= \sum_{m=0}^{\infty} 0.3^m \frac{1 - \left(\frac{0.2}{0.3}\right)^{m+1}}{1 - \frac{0.2}{0.3}} Z_{t-m}.\end{aligned}$$

That is,  $\psi_j = 0.3^j \frac{1 - \left(\frac{0.2}{0.3}\right)^{j+1}}{1 - \frac{0.2}{0.3}}$ .

# Covariance structure of AR( $p$ ) model

Q: What is the relationship between the **covariance function**  $\gamma(k)$  and the **parameters**  $\{\phi_i\}$  of a causal AR( $p$ ) model

$$Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t ?$$

A: **Yule-Walker equations** can be used

- Multiply both sides by  $Y_t$ , then take expectation
- Multiply both sides by  $Y_{t-1}$ , then take expectation
- .....
- Multiply both sides by  $Y_{t-p}$ , then take expectation
- Solve  $p + 1$  equations for  $p + 1$  unknowns for  $(\gamma(0), \gamma(1), \dots, \gamma(p))$

# Yule-Walker equations

$$\text{AR}(p): Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$$

- Multiply both sides by  $Y_t$  and take expectation:

$$\begin{aligned} E(Y_t Y_t) &= E(\phi_1 Y_{t-1} Y_t + \cdots + \phi_p Y_{t-p} Y_t + Z_t Y_t) \\ \Rightarrow \gamma(0) &= \phi_1 \gamma(1) + \cdots + \phi_p \gamma(p) + \sigma^2. \end{aligned}$$

Note that stationarity implies that the covariance function is independent of  $t$ .

- Multiply both sides by  $Y_{t-1}$  and take expectation:

$$\Rightarrow \gamma(1) = \phi_1 \gamma(0) + \phi_2 \gamma(1) + \cdots + \phi_p \gamma(p-1).$$

- In general,

$$\gamma(k) = \phi_1 \gamma(k-1) + \cdots + \phi_p \gamma(k-p), \quad k = 1, 2, \dots$$

Note that causality implies  $E(Y_{t-k} Z_t) = 0$  for  $k = 1, 2, \dots$

# Yule-Walker equations

- We have,

$$\begin{aligned}\gamma(0) &= \phi_1\gamma(1) + \phi_2\gamma(2) + \cdots + \phi_p\gamma(p) + \sigma^2 \\ \gamma(1) &= \phi_1\gamma(0) + \phi_2\gamma(1) + \cdots + \phi_p\gamma(p-1) \\ \gamma(2) &= \phi_1\gamma(1) + \phi_2\gamma(0) + \cdots + \phi_p\gamma(p-2) \\ &\vdots \\ \gamma(p) &= \phi_1\gamma(p-1) + \phi_2\gamma(p-2) + \cdots + \phi_p\gamma(0).\end{aligned}$$

- In matrix form, we have

$$\begin{pmatrix} 1 & -\phi_1 & -\phi_2 & \cdots & -\phi_p \\ -\phi_1 & 1-\phi_2 & -\phi_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\phi_p & -\phi_{p-1} & -\phi_{p-2} & \cdots & 1 \end{pmatrix} \begin{pmatrix} \gamma(0) \\ \gamma(1) \\ \vdots \\ \gamma(p) \end{pmatrix} = \begin{pmatrix} \sigma^2 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

- We can solve for  $\gamma(k)$ s once  $\phi_j$ s are known

## Example: AR(1) Model

Consider a causal AR(1) model,

$$Y_t = \phi Y_{t-1} + Z_t.$$

The Yule-Walker equations yield

$$\begin{aligned}\gamma(0) &= \phi\gamma(1) + \sigma^2, \\ \gamma(k) &= \phi\gamma(k-1), \text{ for } k = 1, 2, \dots\end{aligned}$$

The first equation and the second when  $k = 1$  implies

$$\gamma(0) = \phi^2\gamma(0) + \sigma^2,$$

and thus  $\gamma(0) = \frac{\sigma^2}{1-\phi^2}$ . The ACVF is therefore given by

$$\gamma(k) = \frac{\phi^k \sigma^2}{1 - \phi^2} \text{ for } k = 0, 1, 2, \dots$$

## Example: AR(2) Model

Consider a causal AR(2) model,

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t.$$

The Yule-Walker equations yield

$$\begin{aligned}\gamma(0) &= \phi_1 \gamma(1) + \phi_2 \gamma(2) + \sigma^2, \\ \gamma(1) &= \phi_1 \gamma(0) + \phi_2 \gamma(1), \\ \gamma(k) &= \phi_1 \gamma(k-1) + \phi_2 \gamma(k-2), \text{ for } k = 2, 3, \dots\end{aligned}$$

Divide both sides of the second equation by  $\gamma(0)$  yields

$$\rho(1) = \phi_1 \rho(0) + \phi_2 \rho(1).$$

Since,  $\rho(0) = 1$ , we have  $\rho(1) = \frac{\phi_1}{1-\phi_2}$ .

## Example: AR(2) Model

Take  $k = 2$  in the third equation and divide both sides by  $\gamma(0)$  yields

$$\begin{aligned}\rho(2) &= \phi_1\rho(1) + \phi_2\rho(0) \\ &= \phi_1 \frac{\phi_1}{1 - \phi_2} + \phi_2 \\ &= \frac{\phi_1^2}{1 - \phi_2} + \phi_2.\end{aligned}$$

Substitute  $\gamma(1) = \rho(1)\gamma(0)$  and  $\gamma(2) = \rho(2)\gamma(0)$  into the first equation yields

$$\begin{aligned}\gamma(0) &= \phi_1\rho(1)\gamma(0) + \phi_2\rho(2)\gamma(0) + \sigma^2, \\ \implies \gamma(0) &= \frac{\sigma^2}{1 - \phi_1\rho(1) - \phi_2\rho(2)}.\end{aligned}$$

Successive values of  $\gamma(k)$  and  $\rho(k)$  for  $k \geq 3$  can be obtained recursively.



# Exercise

Find the ACVF of the following AR models with  $Z_t \sim \text{WN}(0, 1)$ .

- a)  $Y_t - 0.4Y_{t-1} = Z_t$
- b)  $Y_t - 0.6Y_{t-1} - 0.27Y_{t-2} = Z_t$
- c)  $Y_t - 0.6Y_{t-1} - 0.72Y_{t-2} = Z_t$  (Is the answer reasonable? Why?)

Solution:

- a) This is an AR(1) model with  $\phi = 0.4$  and  $\sigma^2 = 1$ , we have  
$$\gamma(k) = \frac{0.4^k}{1-0.4^2}, \quad k = 0, 1, 2, \dots$$

# Solution

- b) This is an AR(2) model with  $\phi_1 = 0.6$ ,  $\phi_2 = 0.27$ , and  $\sigma^2 = 1$ , we have

$$\rho(1) = \frac{\phi_1}{1 - \phi_2} = 0.8219$$

$$\rho(2) = \frac{\phi_1^2}{1 - \phi_2} + \phi_2 = 0.7631$$

$$\gamma(0) = \frac{\sigma^2}{1 - \phi_1\rho(1) - \phi_2\rho(2)} = 3.3244$$

$$\gamma(1) = \rho(1)\gamma(0) = 2.7325$$

$$\gamma(2) = \rho(2)\gamma(0) = 2.5371$$

$$\gamma(3) = \phi_1\gamma(2) + \phi_2\gamma(1) = 2.2600$$

$$\gamma(4) = \phi_1\gamma(3) + \phi_2\gamma(2) = 2.0410$$

...

- c) This is an AR(2) model with  $\phi_1 = 0.6$ ,  $\phi_2 = 0.72$ , and  $\sigma^2 = 1$ , we have

$$\rho(1) = \frac{\phi_1}{1 - \phi_2} = 2.1429$$

$$\rho(2) = \frac{\phi_1^2}{1 - \phi_2} + \phi_2 = 2.0057$$

Since  $\rho(1)$  and  $\rho(2)$  are larger than 1, the results are not reasonable. The problem is that this model is not causal.

## Exercise

Consider the process  $Y_t = 0.5Y_{t-1} - 0.06Y_{t-2} + Z_t$ ,  $Z_t \sim WN(0, 1)$ . Find the ACF  $\rho(k)$  for  $k = 0, 1, 2, 3, 4, \dots$

Solution: This is an AR(2) model with  $\phi_1 = 0.5$ ,  $\phi_2 = -0.06$  and  $\sigma^2 = 1$ . We have

$$\rho(0) = 1, \quad \rho(1) = \frac{0.5}{1 - (-0.06)} = 0.472,$$

and for  $k = 2, 3, 4, \dots$

$$\rho(k) = 0.5\rho(k-1) - 0.06\rho(k-2).$$

# Causality condition of AR(2) model

**Q:** For  $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t \Rightarrow (1 - \phi_1 B - \phi_2 B^2) Y_t = Z_t$ ,  
Any conditions for  $\phi_1$  and  $\phi_2$  such that  $Y_t$  is causal?

**A:** i. Causal

$\Rightarrow$  No root for  $\phi(x) = 1 - \phi_1 x - \phi_2 x^2 = 0$  is inside the unit circle

$\Rightarrow \phi(0), \phi(1), \phi(-1)$  are of the same sign

(Otherwise, there exist root(s) between -1 and 1)

$\Rightarrow$  Using  $\phi(0) = 1, \phi(1) = 1 - \phi_1 - \phi_2, \phi(-1) = 1 + \phi_1 - \phi_2$ ,  
we have  $\phi_1 + \phi_2 < 1, \phi_1 - \phi_2 > -1$ .

ii. Causal  $\Rightarrow$  Roots for  $\phi(x) = 1 - \phi_1 x - \phi_2 x^2 = 0$  are greater than 1

$\Rightarrow$  Roots for  $x^2 + \frac{\phi_1}{\phi_2} x - \frac{1}{\phi_2} = 0$  are greater than 1

$\Rightarrow |\text{Product of roots}| = |-\frac{1}{\phi_2}|$  is greater than 1

$\Rightarrow |\frac{1}{\phi_2}| > 1$  or  $|\phi_2| < 1$

# Causality condition of AR(2) model

To conclude, for AR(2):  $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$

The conditions for  $\phi_1$  and  $\phi_2$  such that  $\{Y_t\}$  is causal are:

1.  $\phi_1 + \phi_2 < 1$
2.  $\phi_1 - \phi_2 > -1$
3.  $|\phi_2| < 1$

# Definition of ARMA model

## Definition 15

$\{Y_t\}$  is said to be an ARMA( $p, q$ ) process if

$$\phi(B)Y_t = \theta(B)Z_t, \quad Z_t \sim WN(0, \sigma^2),$$

where

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$$

$$\theta(B) = 1 - \theta_1 B - \dots - \theta_q B^q$$

are the characteristic polynomials with **NO** common roots.

- If there is a common root for  $\phi(B)$  and  $\theta(B)$ , say  $\frac{1}{\xi}$ , then,

$$\begin{aligned} \phi(B)Y_t &= \theta(B)Z_t \\ \Rightarrow \cancel{(1 - \xi B)}\phi^*(B)Y_t &= \cancel{(1 - \xi B)}\theta^*(B)Z_t \end{aligned}$$

That is, the ARMA( $p, q$ ) reduces to an ARMA( $p - 1, q - 1$ ) model.

# Why do we use ARMA models?

- In practice, one often needs to use a high order AR or MA to capture the complex structure of the series. Many parameters are needed.
- Sometimes an ARMA model with very few parameters (e.g. ARMA(1,1), ARMA(2,2)) is enough to achieve the goal
- Models with fewer parameters are preferred



# Stationarity, Causality and Invertibility of ARMA( $p, q$ ) Models

- ARMA( $p, q$ ) model:  $\phi(B)Y_t = \theta(B)Z_t$ ,  $Z_t \sim WN(0, \sigma^2)$ .
- Stationary solution exists if
  - All roots of  $\phi(B)$  are **not on** the unit circle
- Causal if
  - All roots of  $\phi(B)$  are **outside** the unit circle
  - $\Rightarrow Y_t = \phi^{-1}(B)\theta(B)Z_t = \sum_{k \geq 0} \psi_k Z_{t-k}$ , with  $\sum_{k \geq 0} |\psi_k| < \infty$ .
- Invertible if
  - All roots of  $\theta(B)$  are **outside** the unit circle
  - $\Rightarrow Z_t = \theta^{-1}(B)\phi(B)Y_t = \sum_{k \geq 0} \pi_k Y_{t-k}$ , with  $\sum_{k \geq 0} |\pi_k| < \infty$ .

## Example of ARMA( $p, q$ ) Models

$$\text{ARMA}(1,1): Y_t - \phi Y_{t-1} = Z_t - \theta Z_{t-1}, Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

**Q:** Is the model causal/invertible? What is the MA representation  $Y_t = \sum_{j \geq 0} \psi_j Z_{t-j}$ ?

**A:** The model is causal if  $|\phi| < 1$ , invertible if  $|\theta| < 1$ .

Given that  $|\phi| < 1$ , the MA representation is given by

$$(1 - \phi B)Y_t = (1 - \theta B)Z_t$$

$$\Rightarrow Y_t = (1 - \phi B)^{-1}(1 - \theta B)Z_t$$

$$= (1 + \phi B + \phi^2 B^2 + \cdots)(1 - \theta B)Z_t$$

$$= (1 + (\phi - \theta)B + (\phi - \theta)\phi B^2 + (\phi - \theta)\phi^2 B^3 + \cdots)Z_t$$

$$\therefore \psi_0 = 1, \psi_j = (\phi - \theta)\phi^{j-1}, j = 1, 2, \dots$$

ARMA(1,1):  $Y_t - \phi Y_{t-1} = Z_t - \theta Z_{t-1}$ ,  $Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

**Q:** What is the ACF  $\rho(k)$  of the process?

**A: Method 1:** Using MA representation  $Y_t = \sum_{j \geq 0} \psi_j Z_{t-j}$ ,  
 $\psi_0 = 1, \psi_j = (\phi - \theta)\phi^{j-1}$ .

$$\begin{aligned}\gamma(k) &= \text{Cov} \left( \sum_{j \geq 0} \psi_j Z_{t-j}, \sum_{j \geq 0} \psi_j Z_{t+k-j} \right) = \sum_{j=0}^{\infty} \psi_j \psi_{j+k} \sigma^2 \\&= \begin{cases} \psi_0 \psi_k \sigma^2 + \sum_{j=1}^{\infty} \psi_j \psi_{j+k} \sigma^2, & k \neq 0 \\ \psi_0^2 \sigma^2 + \sum_{j=1}^{\infty} \psi_j^2 \sigma^2, & k = 0 \end{cases} \\&= \begin{cases} (\phi - \theta) \phi^{k-1} \sigma^2 + (\phi - \theta)^2 \sum_{j=1}^{\infty} \phi^{2j+k-2} \sigma^2, & k \neq 0 \\ \sigma^2 + (\phi - \theta)^2 \sum_{j=1}^{\infty} \phi^{2j-2} \sigma^2, & k = 0 \end{cases} \\&= \begin{cases} (\phi - \theta) \phi^{k-1} \sigma^2 + (\phi - \theta)^2 \frac{\phi^k}{1 - \phi^2} \sigma^2, & k \neq 0 \\ \sigma^2 + \frac{(\phi - \theta)^2}{1 - \phi^2} \sigma^2, & k = 0 \end{cases}\end{aligned}$$

ARMA(1,1):  $Y_t - \phi Y_{t-1} = Z_t - \theta Z_{t-1}$ ,  $Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

Q: What is the ACF  $\rho(k)$  of the process?

A: **Method 2:** Yule-Walker equation

Multiply  $Y_t$  and  $Y_{t-1}$  on both sides and take expectation respectively

$$\begin{aligned}\gamma(0) - \phi\gamma(1) &= (1 - \phi\theta + \theta^2)\sigma^2 \\ \gamma(1) - \phi\gamma(0) &= -\theta\sigma^2\end{aligned}$$

Solving the two equations gives

$$\gamma(0) = \frac{(1 - 2\phi\theta + \theta^2)\sigma^2}{1 - \phi^2} \text{ and } \gamma(1) = \frac{(\phi - \theta)(1 - \phi\theta)\sigma^2}{1 - \phi^2}$$

$$\text{ARMA}(1,1): Y_t - \phi Y_{t-1} = Z_t - \theta Z_{t-1}, Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

For  $k > 1$ , multiply  $Y_{t-k}$  on both sides and take expectation gives

$$\begin{aligned} \gamma(k) - \phi \gamma(k-1) &= 0 \\ \Rightarrow \gamma(k) &= \phi \gamma(k-1) \\ &= \phi^2 \gamma(k-2) \\ &= \dots \\ &= \phi^{k-1} \gamma(1) \\ &= \phi^{k-1} \frac{(\phi - \theta)(1 - \phi\theta)\sigma^2}{1 - \phi^2}, \end{aligned}$$

which agrees with the answer in Method 1.

## Exercise

Consider the process

$$Y_t = 0.6Y_{t-1} + Z_t - 0.2Z_{t-1}, \quad Z_t \sim WN(0, 4).$$

- (a) Using Yule-Walker equations, find the ACVF and ACF of  $\{Y_t\}$ .
- (b) Find  $\text{Var}(\sum_{t=1}^4 Y_t)$ .

Solution: (a) We have  $\phi = 0.6$ ,  $\theta = 0.2$ , and  $\sigma = 2$ . the ACVF is

$$\gamma(k) = \begin{cases} 5 & k = 0, \\ 2.2 & k = \pm 1, \\ 2.2(0.6)^{|k|-1} & |k| \geq 2. \end{cases}$$

The ACF is  $\rho(k) = \gamma(k)/\gamma(0)$ , which is

$$\rho(k) = \begin{cases} 1 & k = 0, \\ 0.44 & k = \pm 1, \\ 0.44(0.6)^{|k|-1} & |k| \geq 2. \end{cases}$$

(b) By directly counting the number of autocovariance terms with different lags, we have

$$\begin{aligned}\text{Var}\left(\sum_{t=1}^4 Y_t\right) &= 4\gamma(0) + 6\gamma(1) + 4\gamma(2) + 2\gamma(3) \\ &\quad \text{(16 terms in total)} \\ &= 4(5) + 6(2.2) + 4(2.2)0.6 + 2(2.2)0.6^2 \\ &= 40.064.\end{aligned}$$

## Exercise

Consider the model  $Y_t = \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2}$  where  $\alpha, \beta, \theta$  are constants and  $Z_t \sim WN(0, \sigma^2)$ .

- (a) What is the name of this model?
- (b) Give the conditions for  $\{Y_t\}$  to be (i) stationary, (ii) causal (iii) invertible.
- (c) Using (i)  $(1 - x)^{-1} = \sum_{j=0}^{\infty} x^j$  and (ii) first principle, find the stationary solution of  $\{Y_t\}$  for the case  $|\theta| < 1$ .
- (d) Find the stationary solution of  $\{Y_t\}$  for the case  $|\theta| > 1$ .
- (e) Let  $|\theta| < 1$ . Find the variance and autocorrelation function of  $\{Y_t\}$ .



# Solution I

(a) ARMA(1,2) model, given that the characteristic polynomials have no common roots.

(b) (i) The process  $\{Y_t\}$  is stationary if  $|\theta| \neq 1$ . However, if  $|\theta| = 1$ ,  $Y_t$  cannot be represented by  $Z_t$  with decaying coefficients. Thus  $Y_t$  is not stationary.

(ii) The process  $\{Y_t\}$  is causal if  $\phi(x) \neq 0$  for  $|x| \leq 1$ , i.e.,  $|\theta| < 1$ .

(iii) The process  $\{Y_t\}$  is invertible if  $\theta(x) = 1 + \alpha x + \beta x^2 \neq 0$  for  $|x| \leq 1$ .

## Solution II

(c) (i) For  $|\theta| < 1$ , the stationary solution is given by

$$\begin{aligned} Y_t &= \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2} \\ &= (1 - \theta B)^{-1} (1 + \alpha B + \beta B^2) Z_t \\ &= \left( \sum_{j=0}^{\infty} \theta^j B^j \right) (1 + \alpha B + \beta B^2) Z_t \\ &= \left( \sum_{j=0}^{\infty} \theta^j B^j + \sum_{j=0}^{\infty} \alpha \theta^j B^{j+1} + \sum_{j=0}^{\infty} \beta \theta^j B^{j+2} \right) Z_t \\ &= \left( 1 + (\theta + \alpha) B + \sum_{j=2}^{\infty} (\theta^2 + \alpha \theta + \beta) \theta^{j-2} B^j \right) Z_t \\ &= Z_t + (\theta + \alpha) Z_{t-1} + \sum_{j=2}^{\infty} (\theta^2 + \alpha \theta + \beta) \theta^{j-2} Z_{t-j}. \end{aligned}$$

(ii) Using the first principle, we have

$$\begin{aligned} Y_t &= \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2} \\ &= Z_t + (\alpha + \theta)Z_{t-1} + (\beta + \alpha\theta)Z_{t-2} + \beta\theta Z_{t-3} + \theta^2 Y_{t-2} \\ &= Z_t + (\alpha + \theta)Z_{t-1} + (\beta + \alpha\theta + \theta^2)Z_{t-2} + (\alpha\theta^2 + \beta\theta)Z_{t-3} \\ &\quad + \beta\theta^2 Z_{t-4} + \theta^3 Y_{t-3} \\ &= Z_t + (\alpha + \theta)Z_{t-1} + (\beta + \alpha\theta + \theta^2)Z_{t-2} + (\beta + \alpha\theta + \theta^2)\theta Z_{t-3} \\ &\quad + (\beta\theta^2 + \alpha\theta^3)Z_{t-4} + \beta\theta^3 Z_{t-5} + \theta^4 Y_{t-4} \\ &= \dots \\ &= Z_t + (\theta + \alpha)Z_{t-1} + \sum_{i=2}^{\infty} (\theta^2 + \alpha\theta + \beta)\theta^{i-2} Z_{t-i}. \end{aligned}$$

## Solution IV

(d) For  $|\theta| > 1$ , the stationary solution is given by

$$\begin{aligned}Y_t &= \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2} \\Y_{t-1} &= -\left(1 - \frac{1}{\theta} B^{-1}\right)^{-1} \frac{1}{\theta} (1 + \alpha B + \beta B^2) Z_t \\&= -\frac{1}{\theta} \left( \sum_{j=0}^{\infty} \theta^{-j} B^{-j} \right) (1 + \alpha B + \beta B^2) Z_t \\&= -\frac{1}{\theta} \left( \sum_{j=0}^{\infty} \theta^{-j} B^{-j} + \sum_{j=0}^{\infty} \alpha \theta^{-j} B^{1-j} + \sum_{j=0}^{\infty} \beta \theta^{-j} B^{2-j} \right) Z_t \\&= -\left( \frac{\beta B^2}{\theta} + \frac{\beta + \theta \alpha}{\theta^2} B + \sum_{j=0}^{\infty} \frac{\beta + \alpha \theta + \theta^2}{\theta^3} \theta^{-j} B^{-j} \right) Z_t \\Y_t &= -\left( \frac{\beta B^2}{\theta} + \frac{\beta + \theta \alpha}{\theta^2} B + \sum_{j=0}^{\infty} \frac{\beta + \alpha \theta + \theta^2}{\theta^3} \theta^{-j} B^{-j} \right) Z_{t+1} \\&= -\frac{\beta}{\theta} Z_{t-1} - \frac{\beta + \theta \alpha}{\theta^2} Z_t - \sum_{j=0}^{\infty} \frac{\beta + \alpha \theta + \theta^2}{\theta^3} \theta^{-j} Z_{t+1+j} .\end{aligned}$$

## Solution V

(e) From the result

$$Y_t = Z_t + (\theta + \alpha)Z_{t-1} + \sum_{j=2}^{\infty} (\theta^2 + \alpha\theta + \beta)\theta^{j-2}Z_{t-j}$$

in (a), we have

$$\begin{aligned}\text{Var}(Y_t) &= (1 + (\theta + \alpha)^2 + (\theta^2 + \alpha\theta + \beta)^2 + \theta^2(\theta^2 + \alpha\theta + \beta)^2 + \dots) \sigma^2 \\ &= \sigma^2 + (\theta + \alpha)^2 \sigma^2 + \frac{(\theta^2 + \alpha\theta + \beta)^2}{1 - \theta^2} \sigma^2.\end{aligned}$$

The autocovariance function is given by

$$\gamma(k) = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+|k|}$$

where  $\psi_0 = 1$ ,  $\psi_1 = \theta + \alpha$ , and  $\psi_j = \theta^{j-2}(\theta^2 + \alpha\theta + \beta)$ ,  $j \geq 2$ .

# Solution VI

In particular,

$$\begin{aligned}\gamma(1) &= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+1} = \sigma^2 \left( \psi_0 \psi_1 + \psi_1 \psi_2 + \sum_{j=2}^{\infty} \psi_j \psi_{j+1} \right) \\ &= \sigma^2 \left( (\theta + \alpha)(\theta^2 + \alpha\theta + \beta + 1) + \frac{\theta(\theta^2 + \alpha\theta + \beta)^2}{1 - \theta^2} \right),\end{aligned}$$

and for  $k \geq 2$ ,

$$\begin{aligned}\gamma(k) &= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+k} = \sigma^2 \left( \psi_0 \psi_k + \psi_1 \psi_{k+1} + \sum_{j=2}^{\infty} \psi_j \psi_{j+k} \right) \\ &= \sigma^2 \left( (\theta^2 + \alpha\theta + 1)\theta^{k-2}(\theta^2 + \alpha\theta + \beta) + \frac{\theta^k(\theta^2 + \alpha\theta + \beta)^2}{1 - \theta^2} \right).\end{aligned}$$

The autocorrelation function can be obtained by  $\rho_k = \gamma(k)/\gamma(0)$ . Alternatively, Yule-Walker equations can be used to obtain the same answer.

## Definition 16

If  $d$  times of differencing of a process  $\{Y_t\}$ ,  $(1 - B)^d Y_t$ , is a stationary ARMA( $p, q$ ) process, then  $\{Y_t\}$  is an ARIMA( $p, d, q$ ) process, which satisfies

$$\phi(B)(1 - B)^d Y_t = \theta(B)Z_t, \quad Z_t \sim WN(0, \sigma^2),$$

where  $\phi(B)$  and  $\theta(B)$  have no common roots.

- For  $d \geq 1$ , the root of  $(1 - B)^d$  is 1 (on the unit circle), therefore any ARIMA( $p, d, q$ ) process is **non-stationary** and **non-causal**.
- But it is **invertible** if the roots of  $\theta(B)$  are **outside** the unit circle.

# Example 1 of ARIMA Model

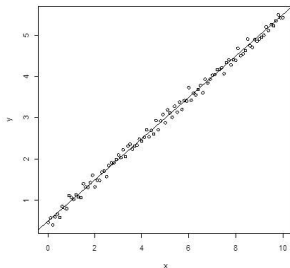
$$\text{Model: } Y_t = \alpha + \beta t + N_t$$

- $\{Y_t\}$  is non-stationary (mean level changes with time)

- 

$$(1 - B)Y_t = \underbrace{\beta + N_t - N_{t-1}}_{\text{MA(1) with unit root}}$$

- $Y_t \sim \text{ARIMA}(0,1,1)$
- $Y_t$  is non-stationary,  
non-causal,  
non-invertible
- $(1 - B)Y_t$  is  
stationary, causal but  
non-invertible

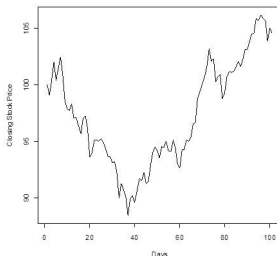




## Example 2 of ARIMA Model

Model (Random Walk):  $Y_t = Y_{t-1} + Z_t$

- Expressed as  $(1 - B)Y_t = Z_t$
- $\{Y_t\} \sim \text{ARIMA}(0,1,0)$
- $\{Y_t\}$  is non-stationary: If  $Y_0 = 0$ ,  $Y_t = Z_t + Z_{t-1} + \cdots + Z_1$   
then  $\text{Var}(Y_t) = t\sigma^2$  **changes with time!**



# What does 'integrated' mean?

- Let  $\{Y_t\}$  be an  $\text{ARIMA}(p,1,q)$  process, then **differencing** the ARIMA  $Y_t$  gives an ARMA  $W_t = (1 - B)Y_t$ .
- In other words, the ARIMA  $Y_t$  is obtained from summing (**integrating**) the ARMA  $W_t$ :

$$\sum_{k=1}^t W_k = \sum_{k=1}^t (Y_k - Y_{k-1}) = Y_t - Y_0 = Y_t \quad \text{if } Y_0 = 0$$

# Stock Price and ARMA model

Let  $P_t$  = Price of stock at the end of date  $t$  and  
 $Y_t = \log P_t = \log(\text{stock price})$

- By Taylor's series expansion of  $f(x) = \log x$  around  $x = 1$ ,

$$\log x = (x - 1) - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3 + \dots$$

$$\begin{aligned} Y_t - Y_{t-1} &= \log \frac{P_t}{P_{t-1}} \\ &= \frac{P_t}{P_{t-1}} - 1 + \text{smaller terms} \\ &\approx \frac{P_t - P_{t-1}}{P_{t-1}} = \text{daily return} \triangleq Z_t \end{aligned}$$

- Common Assumption:

$\log P_t = Y_t$  follow ARIMA(0,1,0) (Random Walk)

Seasonal Component:  $Y_t \approx Y_{t-s} \approx Y_{t-2s} \approx Y_{t-3s} \approx \dots$

## Example

$$Y_t = \phi_1 Y_{t-4} + \phi_2 Y_{t-8} + Z_t$$

can be used to model quarterly data  
(e.g. this summer depends on last summer and the summer before the last one ...)

## Definition 17

$\{Y_t\}$  follows an SARIMA( $p, d, q$ )  $\times$  ( $P, D, Q$ ) $_s$  model if

$$\phi(B)\Phi_P(B^s)(1-B)^d(1-B^s)^DY_t = \theta(B)\Theta_Q(B^s)Z_t$$

where

$$\begin{aligned}\phi(B) &= 1 - \phi_1 B - \dots - \phi_p B^p \\ \theta(B) &= 1 - \theta_1 B - \dots - \theta_q B^q \\ \Phi_P(B^s) &= 1 - \Phi_1 B^s - \dots - \Phi_P B^{sP} \\ \Theta_Q(B^s) &= 1 - \Theta_1 B^s - \dots - \Theta_Q B^{sQ}\end{aligned}$$

Remarks:

- SARIMA can be expressed as high order ARIMA model
- But we love the physical interpretation about seasonal effect

# Exercise

Identify the order of the following Seasonal-ARIMA models

a)  $Y_t - 0.2Y_{t-1} - 0.3Y_{t-4} + 0.06Y_{t-5} = Z_t + 0.6Z_{t-4} - 0.16Z_{t-8}$

b)  $Y_t + 0.2Y_{t-1} - Y_{t-6} - 0.2Y_{t-7} = Z_t + 0.2Z_{t-6}$

c)  $Y_t - Y_{t-1} + 0.2Y_{t-6} - 0.2Y_{t-7} = Z_t + 0.2Z_{t-6}$

Solution:

a)

$$\begin{aligned} Y_t - 0.2Y_{t-1} - 0.3Y_{t-4} + 0.06Y_{t-5} &= Z_t + 0.6Z_{t-4} - 0.16Z_{t-8} \\ (1 - 0.2B - 0.3B^4 + 0.06B^5)Y_t &= (1 + 0.6B^4 - 0.16B^8)Z_t \\ (1 - 0.2B)(1 - 0.3B^4)Y_t &= (1 - 0.2B^4)(1 + 0.8B^4)Z_t. \end{aligned}$$

It is a  $\text{SARIMA}(1, 0, 0) \times (1, 0, 2)_4$  model.

b)

$$\begin{aligned}Y_t + 0.2Y_{t-1} - Y_{t-6} - 0.2Y_{t-7} &= Z_t + 0.2Z_{t-6} \\(1 + 0.2B - B^6 - 0.2B^7)Y_t &= (1 + 0.2B^6)Z_t \\(1 + 0.2B)(1 - B^6)Y_t &= (1 + 0.2B^6)Z_t.\end{aligned}$$

It is a SARIMA(1, 0, 0)  $\times$  (0, 1, 1)<sub>6</sub> model.

c)

$$\begin{aligned}Y_t - Y_{t-1} + 0.2Y_{t-6} - 0.2Y_{t-7} &= Z_t + 0.2Z_{t-6} \\(1 - B + 0.2B^6 - 0.2B^7)Y_t &= (1 + 0.2B^6)Z_t \\(1 - B)(1 + 0.2B^6)Y_t &= (1 + 0.2B^6)Z_t \\(1 - B)Y_t &= Z_t\end{aligned}$$

It is an ARIMA(0, 1, 0) model.

# Summary

- **ARIMA model**:  $\phi(B)(1 - B)^d Y_t = \theta(B)Z_t, Z_t \sim WN(0, \sigma^2)$
- **Stationarity, Causality and Invertibility** -  
Characteristic roots of  $\phi(x) = 0, \theta(x) = 0$  lie outside unit circle
- Find **Causal/MA representation**  $Y_t = \sum_{j \geq 0} \psi_j Z_{t-j}$   
  
Find **AR representation**  $Z_t = \sum_{j \geq 0} \psi_j Y_{t-j}$
- Find **ACF**  $\gamma(k)$   
(M1) Use causal representation  $Y_t = \sum_{j \geq 0} \psi_j Z_{t-j}$   
(M2) Multiply both sides by  $Y_{t-k}, k = 0, 1, \dots, p$  and take expectation, then solve for  $\gamma(k)$ s.
- **SARIMA model**:  
 $\phi(B)\Phi_P(B^s)(1 - B)^d(1 - B^s)^D Y_t = \theta(B)\Theta_Q(B^s)Z_t$